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On the Homology and Cohomology of Congruence Subgroups*

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Introduction

For any prime p, let $\Gamma(n; p)$ denote the congruence subgroup of $SL(n; \mathbf{Z})$ of level p. Then $\Gamma(n; p)$ is the kernel of the surjective mapping

$$SL(n; \mathbf{Z}) \to SL(n; \mathbf{F}_p)$$

induced by reduction mod p. (Here $\mathbf{F}_p$ is the field of p elements.) It follows that the homology and cohomology groups of $\Gamma(n; p)$ are modules over $SL(n; \mathbf{F}_p)$. It is our purpose here to study these modules.

Our results (stated in Section 1) are far from complete. In Section 2, we determine, for arbitrary $n \geqq 3$ and all primes p, the modules $H_1\Gamma(n; p)$. The proof uses the solution to the congruence subgroup problem. The next three sections are devoted to determining $H_N\Gamma(n; p)$ and $H^N\Gamma(n; p)$ for $p=3$ where $N=\frac{1}{2}(n^2-n)$. ($H_q\Gamma(n; p)=H_q\Gamma(n; p)=0$ for $q>N$ follows from the Borel-Serre duality; see the discussion following Theorem 1.2 below.) This involves an explicit free resolution for the Steinberg module $St(n)$ over $SL(n; \mathbf{Z})$ (defined in Section 3) and the fact that $H_0(SL(n; \mathbf{Z}); St(n))=0$ (proved in Section 4).

The remainder of the paper is devoted to the determination of $H_2\Gamma(3; 3)$ and $H^2\Gamma(3; 3)$. These are the most difficult results of the paper requiring an explicit description (given in Section 6) of the boundary of the Borel-Serre compactification of the locally symmetric space

$$SO(3)\backslash SL(n; R)/\Gamma(3; 3).$$

In Section 7, we use this description to compute $H_2\Gamma(3; 3)$ and $H^2\Gamma(3; 3)$ from several key propositions. These propositions are proved in Sections 8 through 12.

Our original motivation for this work was an effort to compute the order of $K_3(Z)$ (see [8]). The results of Quillen (unpublished) and Karoubi [6] show that the order of $K_3(Z)$ must be at least 48. In fact, in [9] we use the results of this paper to show that $K_3(Z)$ is cyclic of order 48.

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1. Statement of Results

In this section, we state the principal results of the paper. Throughout the paper, all homology and cohomology will be with integer coefficients unless another coefficient domain is explicitly indicated.

Let $\mathscr{M}(n; p)$ denote the abelian group of all $n \times n$ matrices with entries in $\mathbf{F}_p$ and $\mathscr{M}_0(n; p)$ the subgroup of matrices with trace zero. (We write $\mathbf{F}_p$ for the field of p elements.) The group $GL(n; \mathbf{F}_p)$ acts on $\mathscr{M}(n; p)$ and $\mathscr{M}_0(n; p)$ by

$$A \cdot g = g^{-1} A g$$

where $A \in \mathscr{M}(n; p)$ and $g \in GL(n; \mathbf{F}_p)$.

Define $\varphi: \Gamma(n; p) \to \mathscr{M}(n; p)$ by

$$\varphi(A) = \frac{1}{p}(A - I) \bmod p.$$

Theorem 1.1. *The mapping $\varphi: \Gamma(n; p) \to \mathscr{M}(n; p)$ is a homomorphism. Furthermore, for any prime p and integer $n \geqq 3$, φ defines an isomorphism of $H_1(\Gamma(n; p))$ onto $\mathscr{M}_0(n; p)$ as modules over $SL(n; \mathbf{F}_p)$.*

The proof of this theorem is given in Section 2. For $p = 3$, an elementary proof is outlined.

For any integer $n \geqq 2$, we let $St(n)$ denote the Steinberg module for the rational vector space of dimension n, $St(n; p)$ the Steinberg module for the vector space of dimension n over $\mathbf{F}_p$. (See Quillen [12] for example.) Then $St(n)$ is a module over $GL(n; \mathbf{Q})$ and $St(n; p)$ is a module over $GL(n; \mathbf{F}_p)$. It is easily checked that $H_q(\Gamma(n; p); St(n))$ is a module over $SL(n; \mathbf{F}_p)$.

Theorem 1.2. *For any integer $n \geqq 3$, $H_0(\Gamma(n; 3); St(n))$ is isomorphic to $St(n; 3)$ as modules over $SL(n; \mathbf{F}_3)$.*

The Borel-Serre duality theorem (see [4], Theorem 11.4.2) states that for any $q \geqq 0$, $n \geqq 2$, and odd prime p,

$$H^q(\Gamma(n; p)) \simeq H_{N-q}(\Gamma(n; p); St(n)).$$

as modules over $SL(n, \mathbf{F}_p)$ where $N = \frac{1}{2}(n^2 - n)$. As a consequence of this and the universal coefficient theorem we have the

Corollary. *For any integer $n \geqq 3$, $H^N(\Gamma(n; 3))$ is isomorphic to $St(n; 3)$ and $H_N(\Gamma(n; 3))$ is isomorphic to* $\mathrm{Hom}(St(n; 3); \mathbf{Z})$ *as modules over $SL(n; \mathbf{F}_3)$.*

The proof of Theorem 1.2 is given in Section 5. It involves the euclidean algorithm and the following result.

Theorem 1.3. *Let Λ be a ring satisfying the euclidean algorithm and $St(\Lambda^n)$ the Steinberg module associated to the free Λ-module Λ^n of rank n. Then, for $n \geqq 3$, $H_0(GL(n; \Lambda); St(\Lambda^n)) = 0$.*

This theorem is proved in Section 4 using an explicit free resolution of $St(\Lambda^n)$ over $GL(n, \Lambda)$. The resolution is described in Section 3.

To state our next result, we let $\bar{P}_1$, $\bar{P}_2$ be two non conjugate maximal parabolic

subgroups in $SL(3;\mathbf{F}_3)$. Let

$$\rho_1, \rho_2 \colon SL(3;\mathbf{F}_3) \to GL(13;\mathbf{Z})$$

be representations obtained by inducing up from the unique nontrivial one dimensional representation

$$\bar{P}_1, \bar{P}_2 \to GL(1;\mathbf{Z}).$$

Let $\mathbf{A}$ and $\mathbf{B}$ be the $SL(3;\mathbf{F}_3)$ modules defined by ρ_1 and ρ_2 respectively.

Theorem 1.4. *There is a short exact sequence*

$$0 \to \mathbf{A} \oplus \mathbf{B} \to H_2(\Gamma(3;3)) \to \mathbf{Z}_2 \to 0$$

of $SL(3;\mathbf{F}_3)$ *modules.*

(Note: The group $\Gamma(n;3)$ is also the kernel of the homomorphism

$$GL(n;\mathbf{Z}) \to GL(n;\mathbf{F}_3)$$

which is surjective since $\mathbf{F}_3^* = \{\pm 1\}$. Thus, $GL(n;\mathbf{F}_3)$ acts on the homology and cohomology of $\Gamma(n;3)$. Furthermore, for n odd, since $GL(n;\mathbf{F}_3)$ is the direct product of $SL(n;\mathbf{F}_3)$ and its center, Theorem 1.4 determines $H_2(\Gamma(3;3))$ as a $GL(3;\mathbf{F}_3)$ module.)

This is the most difficult result of the paper and the proof occupies Sections 6 through 11. One of the basic results used in this proof is an explicit description of the boundary of the Borel-Serre compactification (see [3] or [4]) of $\Gamma(3;3)\backslash SL(3;\mathbf{R})/SO(3)$. This description is given in Section 6.

To describe $H^2(\Gamma(3;3))$, we let J be the subgroup of scalar matrices in $\mathcal{M}(3;3)$. Then $SL(3;\mathbf{F}_3)$ acts trivially on J so $\mathcal{M}(3;3)/J$ becomes an $SL(3;\mathbf{F}_3)$ module in the obvious way.

Theorem 1.5. *There is an exact sequence*

$$0 \to \mathcal{M}(3;3)/J \to H^2(\Gamma(3;3)) \to \mathrm{Hom}(H_2(\Gamma(3;3));\mathbf{Z}) \to 0$$

of $SL(3;\mathbf{F}_3)$ modules.

This result is a consequence of Theorem 1.1, the universal coefficient theorem, and Lemma 12.1.

Whereas this sequence splits over $\mathbf{Z}$, we prove in Section 12 that it does not split over $\mathbf{Z}[SL(3;\mathbf{F}_3)]$.

2. The $SL(n;F_p)$-Module $H_1(\Gamma(n;p))$

We now proceed with the proof of Theorem 1.1. We first present a proof[1] using the solution of the congruence subgroup problem for $SL(n;\mathbf{Z})$ which works for $n \geqq 3$ and all primes p. We then give a sketch of an elementary proof for the case $p=3$.

---

[1] We would like to thank W. Feit, R. Howe, and J-P. Serre for helpful conversations here

Lemma 2.1. *The mapping* $\varphi\colon \Gamma(n;p)\to \mathscr{M}(n;p)$ *defined by*

$$\varphi(X)=\frac{1}{p}(X-I) \bmod p$$

is a homomorphism.

Proof. Given X, Y in $\Gamma(n;p)$, write $X=I+pA$, $Y=I+pB$ so that $\varphi(X)=A \bmod p$ and $\varphi(Y)=B \bmod p$. Then

$$\begin{aligned}\varphi(XY)&=\varphi(I+p(A+B)+p^2AB)\\&=A+B \bmod p\\&=\varphi(X)+\varphi(Y).\end{aligned}$$

Lemma 2.2. *For any* $X\in\Gamma(n;p)$, $\varphi(X)$ *has trace zero.*

Proof. Write $X=I+pA$ and let $\lambda_1,\dots,\lambda_n$ be the eigenvalues of A. These are algebraic integers since A has integral entries. Since $X\in SL(n;\mathbf{Z})$, we have

$$\begin{aligned}1&=\det X=\det(I+pA)\\&=\prod_{i=1}^{n}(1+p\lambda_i)\end{aligned}$$

Thus trace $A=\lambda_1+\cdots+\lambda_n=0 \bmod p$.

It follows from the lemma above that the image of φ is contained in $\mathscr{M}_0(n;p)$. The next lemma tells us that the image of φ is in fact equal to $\mathscr{M}_0(n;p)$.

Lemma 2.3. *The mapping* $\varphi\colon \Gamma(n;p)\to \mathscr{M}_0(n;p)$ *is surjective.*

Proof. Let e_{ij} be the matrix with 1 in the $(i,j)^{\text{th}}$ place and 0 elsewhere and $\bar{e}_{ij}$ the image of e_{ij} in $\mathscr{M}(n;p)$. Set $d_{ij}=e_{ii}-e_{jj}$ and $\bar{d}_{ij}$ the image of d_{ij} in $\mathscr{M}(n;p)$. Then $\bar{e}_{ij}$ and $\bar{d}_{ij}$ for $i\neq j$ generate $\mathscr{M}_0(n;p)$ and, since φ is a homomorphism, it suffices to prove that $\bar{e}_{ij}$ and $\bar{d}_{ij}$, $i\neq j$, are in the image of φ.

Now $\det(I+pe_{ij})=1$ for $i\neq j$ and $\varphi(I+pe_{ij})=\bar{e}_{ij}$ so $\bar{e}_{ij}\in \operatorname{Im}\varphi$ for $i\neq j$. Furthermore, $\det(I+p(e_{ij}-e_{ji}+d_{ij}))=1$ for $i\neq j$ and $\varphi(I+p(e_{ij}-e_{ji}+d_{ij}))=\bar{e}_{ij}-\bar{e}_{ji}+d_{ij}$. It follows that $\bar{d}_{ij}\in \operatorname{Im}\varphi$ for $i\neq j$ and the lemma is proved.

Corollary. *The sequence*

$$1\to\Gamma(n;p^2)\to\Gamma(n;p)\xrightarrow{\varphi}\mathscr{M}_0(n;p)\to 1$$

is exact.

It follows that the proof of Theorem 1.1 will be complete once we show that

$$\Gamma(n;p^2)\subset[\Gamma(n;p),\Gamma(n;p)].$$

Let $E(n)\subset SL(n;\mathbf{Z})$ be the subgroup generated by the elementary matrices $I+te_{ij}$, $t\in\mathbf{Z}$, $i\neq j$, and let $E(n;p)$ be the normal subgroup of $\Gamma(n;p)$ generated by $I+te_{ij}$, $t\equiv 0 \bmod p$, $i\neq j$. According to Corollary 4.2 and Corollary 4.3(b) of Bass, Milnor, Serre [1], we have

$$\Gamma(n;p^2)=E(n;p^2)$$

for $n \geqq 3$. Using the commutator formula

$$[I + t e_{ij}, I + s e_{jk}] = I + s t e_{ik}.$$

we see that

$$E(n; p^2) \subset [E(n; p), E(n; p)]$$

for $n \geqq 3$. Thus

$$\Gamma(n; p^2) = E(n; p^2) \subset [E(n; p), E(n; p)] \subset [\Gamma(n; p), \Gamma(n; p)]$$

for $n \geqq 3$ and Theorem 1.1 is proved.

Remarks. (i) The proof given above holds for congruence subgroups of $SL(n; \Lambda)$ for a more general class of rings Λ. (See Bass, Milnor, Serre [1], pages 94–95.)

(ii) A version of Theorem 1.1 can be proved for a more general class of arithmetic groups by reducing it to a corresponding local problem. (See Borel [2], page 54.)

We now sketch an elementary proof of Theorem 1.1 for the case $p = 3$. The first two lemmas show that $\Gamma(n; 3)$ has a simple set of generators.

Lemma 2.4. *The group $\Gamma(2; 3)$ is generated by the matrices.*

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 \\ 3 & 1 \end{bmatrix}, \quad \begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}.$$

This lemma is well known and not difficult.

Lemma 2.5. *For $n \geqq 3$, the group $\Gamma(n; 3)$ is generated by the matrices $I + 3e_{1n}$, $I + 3e_{n1}$ together with matrices of the form*

$$\begin{bmatrix} & & & 0 \\ & & & \vdots \\ & A & & \\ & & & 0 \\ 0 & \dots & 0 & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{bmatrix}$$

where A is in $\Gamma(n-1; 3)$.

It follows that $\Gamma(n; 3)$ is generated by elementary matrices $I + 3e_{ij}$, $i \neq j$, together with matrices which are essentially in $\Gamma(2, 3)$.

Proof. Let $\gamma = (\gamma_{ij})$ be an element of $\Gamma(n; 3)$

Case 1. $\gamma_{11} = 1$.

We proceed in this case by using row and column operations to eliminate all but the first entry in the first row and column.

Let $\varepsilon, \eta, \delta$ and μ be defined by $\varepsilon = I - \gamma_{1n} e_{1n}$, $\eta = I - \gamma_{n1} e_{n1}$,

$$\delta = \begin{bmatrix} 1 & -\gamma_{12} & \dots & -\gamma_{1n-1} & 0 \\ & \ddots & & & \\ & & & & \\ & & & & 1 \end{bmatrix}, \quad \mu = \begin{bmatrix} 1 & & \\ -\gamma_{21} & \ddots & \\ -\gamma_{n-1,1} & & \\ 0 & & 1 \end{bmatrix}.$$

Then the matrix $\rho=\mu\,\eta\,\gamma\,\delta\,\varepsilon$ is of the form

$$\begin{bmatrix} 1 & 0 & \dots & 0 \\ 0 & & & \\ \vdots & & A & \\ 0 & & & \end{bmatrix}$$

so $\gamma=\eta^{-1}\,\mu^{-1}\,\rho\,\varepsilon^{-1}\,\delta^{-1}$ is a presentation of γ in terms of the prescribed geferators.

Case 2. $\gamma_{11}\neq 1$.

Let $d=$g.c.d.$(\gamma_{11},\dots,\gamma_{1n-1})$ and set $\gamma'_j=d^{-1}\gamma_{1j}$, $1\leqq j\leqq n$. Then g.c.d.$(\gamma'_1,\dots,\gamma'_{n-1})=1$ and we can find α' in $\Gamma(n-1;3)$ so that $(\gamma'_1,\dots,\gamma'_{n-1})\,\alpha'=(1,0,\dots,0)$. (See, for example, Lee [7], page 157.) Define α by

$$\alpha=\begin{bmatrix} & & 0 \\ & \alpha' & \vdots \\ & & 0 \\ 0 & \dots\ 0 & 1 \end{bmatrix}$$

so that

$$\beta=\gamma\,\alpha=\begin{bmatrix} d & 0 & \dots & 0 & \gamma_{1n} \\ & & * & & \end{bmatrix}.$$

Let $\tau=I+3e_{n,n-1}$. Then

$$\beta\,\tau=\begin{bmatrix} d & 0 & \dots & 3\gamma_{1n} & \gamma_{1n} \\ & & * & & \end{bmatrix}.$$

Since g.c.d. $(d,3\gamma_{1n})=1$, we can repeat the above process to make the first row $(1,0,\dots,0,\gamma_{1n})$. This reduces the situation to case 1 and concludes the proof of Lemma 2.5.

Corollary. *For $n\geqq 2$, the group $\Gamma(n;3)$ is generated by n^2-1 elements.*

This is proved easily by induction on n.

Lemma 2.6. *For $n\geqq 3$, each element of the commutator quotient of $\Gamma(n;3)$ has order three.*

Proof. It is immediate from the commutator formula for elementary matrices that the generators $I+3e_{ij}$ have order three in the commutator quotient of $\Gamma(n;3)$ for $n\geqq 3$. Furthermore, conjugation by the matrix

$$\begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

defines an automorphism of $\Gamma(2;3)$ carrying the matrix

$$\begin{bmatrix} 4 & 3 \\ -3 & -2 \end{bmatrix}$$

onto the matrix

$$\begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}.$$

It follows that each of the generators described in Lemma 2.5 have order three in the commutator quotient of $\Gamma(n; 3)$ for $n \geqq 3$.

Theorem 1.1 with $p=3$ now follows from the corollary to Lemma 2.5 and Lemmas 2.1, 2.2, 2.3, and 2.6 above.

3. The Resolution of the Steinberg Module [2]

Let Λ be a principal ideal domain and Λ^n the standard free Λ-module of rank n. We say $(a_1, \ldots, a_n) \in \Lambda^n$ is *unimodular* if the ideal generated by $a_1, \ldots, a_n$ is Λ.

Let $\mathcal{M}(r, s; \Lambda)$ be the set of all $r \times s$ matrices with entries in Λ. Define $\mathcal{S}_q = \mathcal{S}_q(\Lambda^n)$ to be the set of all $A=(a_{ij})$ in $\mathcal{M}(n+q, n; \Lambda)$ with $(a_{i1}, \ldots, a_{in})$ unimodular for each $i=1, \ldots, n+q$ and $\mathcal{P}_q = \mathcal{P}_q(\Lambda^n)$ the subset of all A in $\mathcal{S}_q(\Lambda^n)$ with the property that each $n \times n$ submatrix has determinant zero. If $C(\mathcal{S}_q)$, $C(\mathcal{P}_q)$ denote the free abelian groups generated by $\mathcal{S}_q$, $\mathcal{P}_q$ respectively, we define

$$\partial\colon C(\mathcal{S}_q) \to C(\mathcal{S}_{q-1})$$

by

$$\partial A = \Sigma(-1)^i A_i$$

where $A \in \mathcal{S}_q$ and A_i in $\mathcal{S}_{q-1}$ is obtained from A by deleting the ith row of A. Clearly $\partial^2 = 0$ and $\partial C(\mathcal{P}_q) \subset C(\mathcal{P}_{q-1})$ so we can form the quotient complex

$$C_q(\Lambda^n) = C(\mathcal{S}_q)/C(\mathcal{P}_q).$$

Note that $GL(n, \Lambda)$ acts on $\mathcal{S}_q$ and $\mathcal{P}_q$ by right multiplication so that $C(\mathcal{S}_q)$ and $C(\mathcal{P}_q)$ are $GL(n, \Lambda)$-modules. Then $C_q(\Lambda^n)$ becomes a right $GL(n; \Lambda)$-module in the obvious way.

Let $St(n; \Lambda)$ denote the Steinberg module for Λ^n. (See the proof of Proposition 3.2. for a definition of $St(n; \Lambda)$.)

Theorem 3.1. *There is an epimorphism* $\varphi\colon C_0(\Lambda^n) \to St(n; \Lambda)$ *of* $GL(n; \Lambda)$*-modules so that*

$$\cdots \to C_q(\Lambda^n) \xrightarrow{\partial} C_{q-1}(\Lambda^n) \xrightarrow{\partial} \cdots \xrightarrow{\partial} C_0(\Lambda^n) \xrightarrow{\varphi} St(n; \Lambda) \to 0$$

is a free $GL(n; \Lambda)$*-resolution of* $St(n; \Lambda)$.

Proof. Let K be the simplical complex whose vertices are unimodular elements in Λ^n and whose simplices are all finite nonempty subsets of vertices. We define a subcomplex L in K to consist of those simplices of K all of whose vertices lie in a proper direct summand of Λ^n. Clearly $GL(n; \Lambda)$ acts on K and L.

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[2] We are indebted to Ken Brown for pointing out an error in our original proof and supplying us with the proof given here

Proposition 3.2. *The groups* $H_q(K, L)$ *vanish for* $q \neq n-1$ *and*

$$H_{n-1}(K, L) \cong St(n; \Lambda)$$

as modules over $GL(n; \Lambda)$.

Proof. We begin by recalling the definition of $St(n; \Lambda)$.

Let $\mathbf{F}$ be the quotient field of Λ. Recall that the Tit's building $\mathscr{T}(n; \Lambda)$ is the simplical complex whose vertices correspond to the proper subspaces of $\Lambda^n \otimes_\Lambda \mathbf{F}$ and whose simplices correspond to sequences

$$V_0 \subset V_1 \subset \cdots \subset V_q$$

of proper subspaces of $\Lambda^n \otimes_\Lambda \mathbf{F}$. It is known that $\mathscr{T}(n; \Lambda)$ has the homotopy type of a wedge of $(n-2)$-spheres (see Quillen [12], page 202–203 or Lusztig [11]; page 12 for example) and, by definition, the Steinberg module for Λ^n is given by

$$St(n; \Lambda) \cong H_{n-2}(\mathscr{T}(n; \Lambda)).$$

Since $GL(n; \Lambda)$ acts on $\mathscr{T}(n; \Lambda)$, $St(n; \Lambda)$ is a $GL(n; \Lambda)$-module.

By the definition of K, it is clearly acyclic so that

$$H_q(K, L) \simeq \tilde{H}_{q-1}(L)$$

for $q > 0$ from the exact sequence of the pair (K, L). Therefore, to prove Proposition 3.2, we need only show that the complex L has the same homology as $\mathscr{T}(n; \Lambda)$.

Let $\{H_i, i \in I\}$ be the set of direct summands in Λ^n of rank $(n-1)$. For each i, we define a subcomplex K_i of L to consist of those simplices all of whose vertices lie in H_i. Since Λ is a principal ideal domain, each H_i is isomorphic to Λ^{n-1}. Identifying H_i with Λ^{n-1} under such an isomorphism, the complex K_i has as vertices all unimodular elements of Λ^{n-1} and as simplices, all finite subsets of vertices. Thus each K_i is contractible.

More generally, if $K_{i_0} \cap \cdots \cap K_{i_q} \neq \emptyset$, then $H_{i_0} \cap \cdots \cap H_{i_q} = H \neq \{0\}$ by definition. Again H is a direct summand in Λ^n isomorphic to Λ^r for some r, $0 < r < n$, and, as above, $K_{i_0} \cap \cdots \cap K_{i_q}$ can be described as the simplical complex whose vertices are all unimodular elements of Λ^r and whose simplices are all finite subsets of vertices. Thus $K_{i_0} \cap \cdots \cap K_{i_q}$ is acyclic.

Let N be the nerve of the covering $\{K_i, i \in I\}$ of L. By the remarks above, this is an acyclic covering so we have

$$H_q(L) \simeq H_q(N)$$

for all $q \geqq 0$. (See for example Lusztig [11], page 6.)

Let $\{W_j, j \in J\}$ be the set of hyperplanes in $\mathbf{F}^n \simeq \Lambda^n \otimes_\Lambda \mathbf{F}$ and let T_j be the subcomplex of $\mathscr{T}(n; \Lambda)$ consisting of all simplices having W_j as a vertex i.e., the star of the vertex W_j. This is easily seen to be an acyclic covering (see Lusztig [11], page 12) so again, if $\tilde{N}$ is the nerve of the covering, we have

$$H_q(\mathscr{T}(n; \Lambda)) \cong H_q(\tilde{N})$$

for all $q \geqq 0$.

Finally, the mapping

$$H \mapsto H \otimes_\Lambda \mathbf{F}$$

from the set of direct summands of Λ^n to the set of subspaces of $\mathbf{F}^n \simeq \Lambda^n \otimes_\Lambda \mathbf{F}$ defines a simplical isomorphism of N onto $\tilde{N}$. Thus we have the sequence of isomorphisms

$$\begin{aligned} H_q(K, L) &\simeq H_{q-1}(L) \\ &\simeq H_{q-1}(N) \\ &\simeq H_{q-1}(\tilde{N}) \\ &\simeq H_{q-1}(\mathscr{T}(n; \Lambda)). \end{aligned}$$

Since each of these isomorphisms is easily seen to preserve the $GL(n; \Lambda)$-action, the proof of Proposition (3.2) is complete.

To prove Theorem 3.1, we note that the $(n-2)$-skeleton of L coincides with the $(n-2)$-skeleton of K so that

$$C_q(K, L) = 0$$

for $q \leqq n-2$. This fact and Proposition 3.2 gives the exact sequence

$$\cdots \to C_{q+n}(K, L) \to C_{q+n-1}(K, L) \to \cdots \to C_{n-1}(K, L) \to St(n; \Lambda) \to 0.$$

Furthermore, for $q \geqq 0$, we have $GL(n; \Lambda)$-chain isomorphisms

$$C_{q+n-1}(K) \simeq C(\mathscr{S}_q),$$

$$C_{q+n-1}(L) \simeq C(\mathscr{P}_q),$$

where $C_*(K)$ and $C_*(L)$ are the ordered simplical chain complexes. It follows that

$$C_q(\Lambda^n) \simeq C_{q+n-1}(K, L)$$

as $GL(n; \Lambda)$-modules so we have the resolution

$$\cdots \to C_{q+1}(\Lambda^n) \to C_q(\Lambda^n) \to \cdots \to C_0(\Lambda^n) \to St(n; \Lambda) \to 0.$$

To see that this is a $GL(n; \Lambda)$-free resolution of $St(n; \Lambda)$, we observe that $C_q(\Lambda^n)$ can be identified with the free abelian group generated by the set of matrices A in $\mathscr{M}(n+q, n; \Lambda)$ satisfying

(i) each row of A is unimodular,
(ii) some $n \times n$-submatrix of A has non-vanishing determinant.

It is immediate that $GL(n; \Lambda)$ acts freely on this set of generators so, picking one element from each orbit, we obtain free $GL(n; \Lambda)$ generators for $C_q(\Lambda^n)$. This completes the proof of Theorem 3.1.

4. The Vanishing of $H_0(SL(n; \Lambda); St(n; \Lambda))$

In theory, one should be able to use the resolution of the previous section to compute the groups $H_q(SL(n; \Lambda); St(n; \Lambda))$ for $q \geqq 0$. However, because of the size of the groups $C_q(\Lambda^n)$, this is impractical except when $q = 0$. In this case we have the following.

Theorem 4.1. *Let Λ be a Euclidean domain with multiplicative norm. Then*

$$H_0(SL(n;\Lambda);\ St(n;\Lambda))=0.$$

Recall that a Euclidean domain Λ has multiplicative norm if $|\lambda\mu|=|\lambda|\cdot|\mu|$ for all λ, μ in Λ. As a consequence, λ is a unit if and only if $|\lambda|=1$.

As an immediate consequence of Theorem 4.1, we have

Corollary. *Let Λ be a Euclidean domain with multiplicative norm. Then*

$$H_0(GL(n;\Lambda);\ St(n;\Lambda))=0.$$

We proceed with the proof. As is well known,

$$H_0(SL(n;\Lambda);\ St(n;\Lambda))=St(n;\Lambda)/K$$

where K is the subgroup of $St(n;\Lambda)$ generated by elements of the form $x-xg$, $x\in St(n;\Lambda)$ and $g\in SL(n;\Lambda)$. Let $\alpha\colon C_0(\Lambda^n)\to St(n;\Lambda)/K$ be the composite

$$C_0(\Lambda^n)\xrightarrow{\varphi} St(n;\Lambda)\to St(n;\Lambda)/K$$

and $\beta\colon GL(n;\Lambda)\to C_0(\Lambda^n)$ the obvious homomorphism.

Lemma 4.2. *For any $g\in GL(n;\Lambda)$, $\alpha\beta(g)=0$.*

Proof. First note that, for any $x\in C_0(\Lambda^n)$, we have $\alpha(xg)=\alpha(x)$ for $g\in SL(n;\Lambda)$. In particular, $\alpha\beta(g)=\alpha\beta(g')$ whenever $\det g=\det g'$. Thus it is enough to prove $\alpha\beta(g)=0$ when g has the form

$$g=\begin{bmatrix}1 & & & & \\ & 1 & & & \\ & & \ddots & & \\ & & & 1 & \\ & & & & a\end{bmatrix}$$

where a is a unit in Λ. Now

$$\partial\begin{bmatrix}1 & & & & & \\ & 1 & & & & \\ -1 & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & & a\end{bmatrix}=\begin{bmatrix} & 1 & & & & \\ -1 & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & a & \end{bmatrix}-\begin{bmatrix} 1 & & & & & \\ -1 & 1 & & & & \\ & & 1 & & & \\ & & & \ddots & & \\ & & & & a & \end{bmatrix}+g$$

$$=g_1-g_2+g.$$

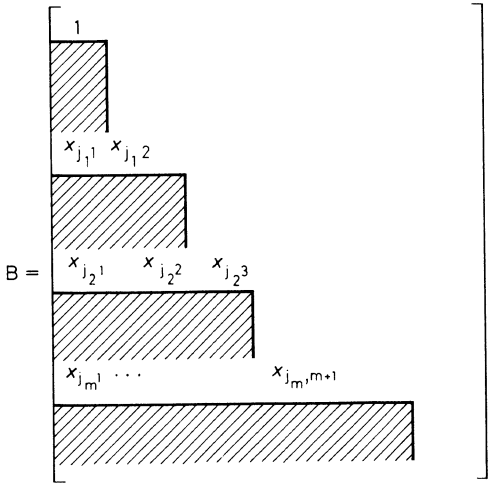
Since $\alpha(\partial y)=0$ for any $y\in C_1(\Lambda^n)$, and $g_1, g_2\in GL(n;\Lambda)$ with $\det g_1=\det g_2=\det g$, we see that

$$\begin{aligned}0&=\alpha(g_1-g_2+g)\\ &=\alpha\beta(g_1)-\alpha\beta(g_2)+\alpha\beta(g)\\ &=\alpha\beta(g).\end{aligned}$$

This proves Lemma 4.2.

Before proceeding, we need the

Definition. Let Λ be a Euclidean domain. A matrix B in $\mathcal{M}(r, s; \Lambda)$ is said to be in lower echelon form if



where $1 < j_1 < \cdots < j_m < s$ and $|x_{j_k, k}| < |x_{j_i, i+1}|$ for $1 \leqq k \leqq i$, $1 \leqq i \leqq m$.

In the above, the unshaded area consists of entries which are zero and the shaded area consists of arbitrary entries.

The following lemma is an easy consequence of the Euclidean algorithm. We omit the proof.

Lemma 4.3. *Let Λ be a Euclidean domain with multiplicative norm. Then, for any matrix A in $S_q(\Lambda^n)$, $q \geqq 0$, $n \geqq 2$, there is a matrix B in $\mathcal{S}_q(\Lambda)$ in lower echelon form such that $B = Ag$ for some g in $SL(n, \Lambda)$.*

We can now prove

Lemma 4.4. *Let Λ be a Euclidean domain with multiplicative norm. Then any element $x \in C_0(\Lambda^n)$ can be written*

$$x = \sum_{j=1}^{r} a_j \beta(g_j) + \partial y$$

for $g_j \in GL(n, \Lambda)$, $1 \leqq j \leqq r$, and $y \in C_1(\Lambda^n)$.

Since α: $C_0(\Lambda^n) \to H_0(SL(n; \Lambda); St(n; \Lambda))$ is surjective and $\alpha\beta = \alpha\partial = 0$, Theorem 4.1 is an immediate consequence of Lemma 4.4.

Proof of Lemma 4.4. We proceed by induction on the norm of the determinant.

Let A be a generator of $C_0(\Lambda^k)$. Then $|\det A| > 0$ (for otherwise A would be 0 in $C_0(\Lambda^k)$) and $|\det A| = 1$ if and only if $A = \beta(g)$ for $g \in GL(n; \Lambda)$. Suppose that $|\det A| = k > 1$ and, by induction, that the lemma is true whenever the norm of the

determinant is less than k. Pick g in $SL(n;\Lambda)$ so that Ag has lower echelon form,

$$Ag=\begin{bmatrix} 1 & & & & \\ x_{21} & x_{22} & & & \\ \cdots\cdots & \cdots\cdots & x_{33} & & \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & \ddots & \\ & & & & x_{nn} \end{bmatrix}.$$

If $|x_{22}|>1$, we consider the equation

$$\partial\begin{bmatrix} 1 & & & \\ x_{21} & x_{22} & & \\ & 1 & & \\ x_{31} & x_{32} & x_{33} & \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & x_{nn} \end{bmatrix}=\begin{bmatrix} x_{21} & x_{22} & & \\ & 1 & & \\ x_{31} & x_{32} & x_{33} & \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & x_{nn} \end{bmatrix}-\begin{bmatrix} 1 & & & \\ & 1 & & \\ x_{31} & x_{32} & x_{33} & \\ \cdots\cdots & \cdots\cdots & \cdots\cdots & x_{nn} \end{bmatrix}+Ag.$$

It follows that, mod $\partial C_1(\Lambda^k)$, Ag is the sum of two terms, each of which has determinant of norm less than k so, by induction, Ag can be written in the required form.

More generally, suppose $|x_{jj}|=1$ for $j<l$ and $|x_{ll}|>1$. Then $x_{ij}=0$ for $i<j<l$. Let $B=(b_{ij})$ be the element of $\mathcal{M}(n+1),n;\Lambda)$ obtained from Ag by

$$b_{ij}=\begin{cases} x_{ij}, & i\leqq l, \\ \delta_{l,j}, & i=l+1, \\ x_{i-1,j}, & i>l+1, \end{cases}$$

where $Ag=(x_{ij})$ as above. Then

$$\partial B=C_1-C_2+\cdots\pm C_{l-1}\pm Ag$$

where $|\det C_j|<k$, $1\leqq j<l$. As above, it follows by induction that Ag has the required form. This completes the proof of Lemma 4.4 and, as remarked above, the proof of Theorem 4.1.

5. The $GL(n;F_3)$ Module $H_0(\Gamma(n;3);St(n))$

For convenience in this section we introduce the following notation

G_n for $GL(n;\mathbf{Z})$,

$\bar{G}_n$ for $GL(n;\mathbf{F}_3)$,

Γ_n for $\Gamma(n;3)$,

and

$\bar{St}(n)$ for $St(n;3)$.

Let $\rho\colon \mathbf{Z}[G_n]\to St(n)$, $\bar{\rho}\colon \mathbf{Z}[\bar{G}_n]\to \bar{St}(n)$ be the composites

$$\mathbf{Z}[G_n]\subset C_0(\mathbf{Z}^n)\xrightarrow{\varphi} St(n)$$

$$\mathbf{Z}[\bar{G}_n]=C_0(\mathbf{F}_3^n)\xrightarrow{\bar{\varphi}}\bar{St}(n).$$

Both of these maps are epimorphisms, ρ as a consequence of Lemma 4.4 and $\bar{\rho}$ because $\bar{\varphi}$ is an epimorphism. Let $R = \ker \rho$, $\bar{R} = \ker \bar{\rho}$. We then have a diagram of exact sequences

$$\begin{array}{ccccccccc} 0 & \longrightarrow & R & \longrightarrow & \mathbf{Z}[G_n] & \longrightarrow & St(n) & \longrightarrow & 0 \\ & & \downarrow & & \downarrow & & \downarrow & & \\ 0 & \longrightarrow & \bar{R} & \longrightarrow & \mathbf{Z}[\bar{G}_n] & \longrightarrow & \bar{St}(n) & \longrightarrow & 0 \end{array}$$

where the vertical mappings are induced by reduction mod 3.

Now Γ_n acts on both sequences, trivially on the second, so, applying the functor $M \mapsto H_*(\Gamma_n; M)$, we obtain the diagram of exact sequences.

$$\begin{array}{ccccccccc} H_1(\Gamma_n; St(n)) & \longrightarrow & H_0(\Gamma_n; R) & \longrightarrow & H_0(\Gamma_n; \mathbf{Z}[G_n]) & \longrightarrow & H_0(\Gamma_n; St(n)) & \longrightarrow & 0 \\ \downarrow & & \downarrow & & \downarrow & & \downarrow & & \\ H_1(\Gamma_n; \bar{St}(n)) & \longrightarrow & H_0(\Gamma_n; \bar{R}) & \longrightarrow & H_0(\Gamma_n; \mathbf{Z}[\bar{G}_n]) & \longrightarrow & H_0(\Gamma_n; \bar{St}(n)) & \longrightarrow & 0. \end{array}$$

Since $H_0(\Gamma_n; M) \simeq M$ if Γ_n acts trivially on M and $H_0(\Gamma_n; \mathbf{Z}[G_n]) \simeq \mathbf{Z}[G_n/\Gamma_n] \simeq \mathbf{Z}[\bar{G}_n]$, this diagram reduces to

$$\begin{array}{ccccccccc} H_1(\Gamma_n; St(n)) & \to & H_0(\Gamma_n; R) & \xrightarrow{\sigma} & \mathbf{Z}[\bar{G}_n] & \xrightarrow{\rho^*} & H_0(\Gamma_n; St(n)) & \to & 0 \\ & & \downarrow{\scriptstyle \alpha} & & \downarrow{\scriptstyle \beta} & & \downarrow{\scriptstyle \gamma} & & \\ 0 & \longrightarrow & \bar{R} & \xrightarrow{\bar{\sigma}} & \mathbf{Z}[\bar{G}_n] & \xrightarrow{\bar{\rho}} & \bar{St}(n) & \to & 0. \end{array} \tag{5.1}$$

where β is the identity map

Define $\delta\colon \bar{R} \to H_0(\Gamma_n; St(n))$ to be the composite $\rho_* \beta^{-1} \bar{\sigma}$.

Lemma 5.1. *The sequence*

$$H_0(\Gamma_n; R) \xrightarrow{\alpha} \bar{R} \xrightarrow{\delta} H_0(\Gamma_n; St(n)) \xrightarrow{\gamma} \bar{St}(n) \to 0$$

is exact.

Proof. This is straightforward. We check only the fact that $\ker \delta \subset \operatorname{Im} \alpha$.

Suppose $\delta a = 0$ for $a \in \bar{R}$. Then $\rho_* \beta^{-1} \bar{\sigma} a = 0$ so, from diagram (5.1), there is an element b in $H_0(\Gamma_n; R)$ such that $\sigma b = \beta^{-1} \bar{\sigma} a$. But then

$$\begin{aligned} \bar{\sigma} \alpha(b) &= \beta \sigma(b) \\ &= \beta(\beta^{-1} \bar{\sigma} a) \\ &= \bar{\sigma} a \end{aligned}$$

so, since $\bar{\sigma}$ is injective $\alpha(b) = a$.

Theorem 1.2 is now a consequence of the following.

Lemma 5.2. *The mapping $\alpha\colon H_0(\Gamma_n; R) \to \bar{R}$ is surjective.*

Proof. Let $\partial\colon C_1(\mathbf{F}_3^n) \to C_0(\mathbf{F}_3^n) \simeq \mathbf{Z}[\bar{G}_n]$ be the boundary mapping of the resolution defined in Section 3. Then $\bar{R} = \operatorname{Im} \partial$. Now, Lemma 4.3 implies that $C_1(\mathbf{F}_3^n)$ is

generated as a $\mathbf{Z}[G_n]$ module by matrices of the form

$$A = \begin{bmatrix} 1 & & & & & & \\ & 1 & & & & & \\ & & \ddots & & & & \\ & & & 1 & & & \\ a_1 & \dots & & a_j & & & \\ & & & & 1 & & \\ & & & & & \ddots & \\ & & & & & & 1 \end{bmatrix}$$

for $1 \leqq j \leqq n$, $a_1, \dots, a_j = 0, \pm 1$. It follows that Im ∂ is generated by elements of the form ∂A. Furthermore, it is easily seen that each term in the expression for ∂A either has determinent ± 1 or is 0 in $C_0(\mathbf{F}_3^n)$ and, each of the terms with determinent ± 1 can be considered as an element of G_n. Thus each element in $\bar{R}$ is the image of an element in R and thus in the image of α (since $H_0(\Gamma_n; R) \simeq R/J$ where J is generated by $x - xg$, $x \in R$, $g \in \Gamma_n$).

This completes the proof of Lemma 5.2.

Remark. The proof above relies heavily on the fact that the only units in $\mathbf{F}_3$ are ± 1. In fact, for primes $p > 3$, it is *not* true that $H_0(\Gamma(n; p)\colon St(n))$ is isomorphic to $St(n\colon p)$. We conjecture that $H_0(\Gamma(n; p); St(n))$ is isomorphic to the $SL(n; \mathbf{F}_p)$-module

$$H_{n-2}(\mathscr{T}(n; Z)/\Gamma(n; p)).$$

6. Blowing up the Building[3]

Let G be a real semisimple $\mathbf{Q}$-group, K a maximal compact subgroup of G, and Γ an arithmetic subgroup of G containing no elements of finite order. Then Γ operates properly discontinuously on the symmetric space $X = K\backslash G$ and X/Γ is a $K(\Gamma; 1)$-manifold with finite volume. Let $M = \bar{X}/\Gamma$ be the compactification of X/Γ defined by Borel and Serre in [3, 4]. Then M is a manifold with boundary and X/Γ is the interior of M. Thus $H_*(\Gamma) \simeq H_*(M)$. In this section, we give an explicit description to ∂M when $G = SL(3; \mathbf{R})$, $K = SO(3)$, and $\Gamma = \Gamma(3; 3)$. This description will be used in the remaining sections to compute $H_2(M) \simeq H_2(\Gamma(3; 3))$.

Recall that the compactification of Borel-Serre referred to above involves first enlarging X to a manifold $\bar{X}$ with corners. The manifold $\bar{X}$ is the disjoint union of cells $e(P)$, one for each rational parabolic subgroup P of G. Furthermore, the group $G(\mathbf{Q})$ of rational points of G acts on $\bar{X}$ with $P(\mathbf{Q})$ taking $e(P)$ into itself. If Γ is a torsion free arithmetic subgroup of $G(\mathbf{Q})$, then Γ acts freely on $\bar{X}$ and the quotient manifold $\bar{X}/\Gamma$ is the Borel-Serre compactification of X/Γ.

[3] We are indebted to Howard Garland for the title here

For the remainder of the paper, we write G for $SL(3, \mathbf{R})$, X for $SO(3)\backslash SL(3, \mathbf{R})$, and Γ for $\Gamma(3; 3)$. Note that $\Gamma(3, 3)$ has no element of finite order.

Let l_0 be the line in $\mathbf{Q}^3$ through the point $(1, 0, 0)$ and h_0 the plane containing l_0 and the point $(0, 1, 0)$. (All lines and planes in this paper contain the origin.) Let $P_1 \subset G$ be the parabolic subgroup leaving l_0 invariant, $P_2 \subset G$ the parabolic subgroup leaving h_0 invariant and $P_0 = P_1 \cap P_2$. For $i = 0, 1, 2$, set $e_i = e(P_i)$ and define spaces Y_i by

$$Y_i = \bar{e}_i \times_{P_i(\mathbf{Q})} G(\mathbf{Q}).$$

Since $\bar{e}_0 \subset \bar{e}_1$, $\bar{e}_0 \subset \bar{e}_2$, we have $Y_0 \subset Y_1$, $Y_0 \subset Y_2$ and we can form the space

$$Y = \underset{Y_0}{Y_1 \cup Y_2}.$$

Define a mapping $\varphi\colon e_i \times G(\mathbf{Q}) \to \bar{X}$ by $\varphi(x, g) = x\,g$ (considering $e_i = e(P_i)$ as a subset of $\bar{X}$). For $h \in P_i(\mathbf{Q})$,

$$\begin{aligned}\varphi_i((x, g)\,h) &= \varphi_i(x\,h, h^{-1}\,g)\\ &= \varphi_i(x, g)\end{aligned}$$

so φ_i defines a mapping $\varphi_i\colon Y_i \to \bar{X}$, $i = 0, 1, 2$.

Lemma 6.1. *For each $i = 0, 1, 2$, the mapping $\varphi_i\colon Y_i \to \bar{X}$ is injective.*

Proof. Let $[x, g]$ denote the element in Y_i determined by (x, g) in $e_i \times G(\mathbf{Q})$ and suppose $\varphi_i[x, g] = \varphi_i[y, h]$. Then $x\,g = y\,h$ so $x\,g\,h^{-1} = y$. But then $g\,h^{-1} \in P_i(\mathbf{Q})$ and

$$[x, g] = [x\,g\,h^{-1}, h\,g^{-1}\,g] = [y, h].$$

Now, every proper $\mathbf{Q}$-parabolic subgroup of $G = SL(3, \mathbf{R})$ is conjugate to one of P_0, P_1, or P_2 so it follows that

$$\partial \bar{X} = \varphi(Y_0) \cup \varphi(Y_1) \cup \varphi(Y_2).$$

Furthermore, since $\bar{e}_0 = \bar{e}_1 \cap \bar{e}_2$, it follows easily that

$$\varphi(Y_0) = \varphi(Y_1) \cap \varphi(Y_2).$$

Thus we have

Lemma 6.2. *The mappings φ_i, $i = 0, 1, 2$, define a diffeomorphism*

$$\varphi\colon Y \to \partial \bar{X}.$$

If we define an action of $SL(3, \mathbf{Q})$ on Y by setting $[x, g]\,h = [x, g\,h]$, it is easily seen that φ is $SL(3, \mathbf{Q})$-equivariant.

Now we define spaces Z_j, $j = 0, 1, 2$, by

$$Z_j = \bar{e}_j \times_{P_j(\mathbf{Z})} G(\mathbf{Z})$$

and let

$$Z = \underset{Z_0}{Z_1 \cup Z_2}.$$

The inclusion $G(\mathbf{Z})\subset G(\mathbf{Q})$ defines mappings $Z_j\to Y_j$ and a mapping $\psi\colon Z\to Y$, each of which is equivariant relative to the action of $G(\mathbf{Z})=SL(3;\mathbf{Z})$.

Lemma 6.3. *The mapping $\psi\colon Z\to Y$ is a diffeomorphism equivariant relative to the action of $SL(3;\mathbf{Z})$.*

Proof. Consider the case $j=1$ and note that we can identify $G(\mathbf{Z})/P_1(\mathbf{Z})$ with the set of lines in $\mathbf{Z}^3$ and $G(\mathbf{Q})/P_1(\mathbf{Q})$ with the set of lines in $\mathbf{Q}^3$. Since $\mathbf{Q}^3$ and $\mathbf{Z}^3$ have the same set of lines, we see that the inclusion $G(\mathbf{Z})\subset G(\mathbf{Q})$ defines a bijection of $G(\mathbf{Z})/P_1(\mathbf{Z})$ onto $G(\mathbf{Q})/P_1(\mathbf{Q})$.

A similar argument holds for $j=0,2$.

It follows from the results above that $\partial\bar{X}/\Gamma$ is diffeomorphic to Z/Γ where $\Gamma=\Gamma(3;3)\subset SL(3;\mathbf{Z})$. By the definition of the manifold Z, we have

Lemma 6.4. *The manifold $\partial\bar{X}/\Gamma$ is diffeomorphic to the manifold*

$$(Z_1/\Gamma)\ \cup\ (Z_2/\Gamma)$$
$$(Z_0/\Gamma)$$

where $\Gamma=\Gamma(3;3)$.

Let $K_j=\bar{e}_j/\Gamma\cap P_j$, $j=0,1,2$. Then K_j is a manifold of type $K(\pi,1)$ with $\pi=\Gamma\cap P_j$. Now, the quotient group $\bar{P}_j=\Gamma\cap P_j(\mathbf{Z})\backslash P_j(\mathbf{Z})$ acts on K_j and on $\Gamma\backslash SL(3;\mathbf{Z})\simeq SL(3;\mathbf{F}_3)$, $j=0,1,2$, and we define manifolds W_j, $j=0,1,2$, by

$$W_j=K_j\times_{\bar{P}_j}SL(3;\mathbf{F}_3).$$

Note that W_1 consists of 13 copies of K_1, one for each line l in $\mathbf{F}_3^3$, W_2 consists of 13 copies of K_2, one for each plane h in $\mathbf{F}_3^3$, and W_0 consists of 52 copies of K_0, one for each pair (l,h) with $l\subset h$. This follows from the fact that the groups $\bar{P}_j$, $j=0,1,2$, are parabolic subgroups of $SL(3;\mathbf{F}_3)$.

Remark. The description of the manifolds, W_0, W_1, and W_2 given above depends on the fact that we are dealing with the prime 3. For primes $p>3$, the manifold $\partial\bar{X}/\Gamma(n;p)$ is obtained by blowing up the complex $\mathscr{T}(n;p)/\Gamma(n;p)$.

Proposition 6.5. *The manifold $\partial\bar{X}/\Gamma$ is diffeomorphic to the manifold.*

$$W_1\cup W_2.$$
$$W_0$$

Proof. By Lemma 6.4, it suffices to prove that Z_j/Γ is diffeomorphic to W_j, $j=0,1,2$. Define mappings

$$\rho\colon \bar{e}_j\times SL(3;\mathbf{Z})\to Z_j/\Gamma$$

$$\sigma\colon \bar{e}_j\times SL(3;\mathbf{Z})\to W_j$$

in the obvious way. Then $\rho(x_1,g_1)=\rho(x_2,g_2)$ if and only if

$$x_1 p\alpha=x_2 \quad\text{and}\quad \beta g_1 p=g_2$$

for some $p\in P_j(\mathbf{Z})$, $\alpha\in P_j(\mathbf{Z})\cap\Gamma$, $\beta\in\Gamma$. Similarly, $\sigma(x_1,g_1)=\sigma(x_2,g_2)$ if and only if

$$x_1 q=x_2 \quad\text{and}\quad \gamma g_1 q=g_2$$

for some $q \in P_j(\mathbf{Z})$, $\gamma \in \Gamma$. It follows easily that $\rho(x_1, g_1) = \rho(x_2, g_2)$ implies $\sigma(x_1, g_1) = \sigma(x_2, g_2)$. To see the converse, set

$$\gamma = \beta g_1 p \alpha^{-1} p^{-1} g_1^{-1}.$$

As we will see below, the four dimensional manifolds K_1, K_2 have four boundary components each diffeomorphic to K_0. It follows from Proposition 6.5 above that $\partial \bar{X}/\Gamma$ consists of 13 copies of K_1, one for each line l in $\mathbf{F}_3^3$ and 13 copies of K_2, one for each plane h in $\mathbf{F}_3^3$. Furthermore, a copy of K_1 corresponding to a line l has a boundary component in common with a copy of K_2 corresponding to a plane h if and only if $l \subset h$. Thus $\partial \bar{X}/\Gamma$ is obtained by „blowing up" the Tits building for $\mathbf{F}_3^3$.

We now describe the manifolds K_0, K_1, and K_2.

Let $X_0 = SO(2) \backslash SL(2, \mathbf{R})$ and $\Gamma_0 = \Gamma(2; 3)$.

Then $M_0 = X_0/\Gamma_0$ is an open surface with $\pi_1 M_0$ a free group on three generators. (See Lemma 2.4.) Thus M_0 can be identified with

$$S^2 - \bigcup_{j=1}^{4} D_j$$

where S^2 is the two sphere and D_1, D_2, D_3, D_4 are disjoint closed disks in S^2. Let $\Gamma_0 \subset SL(2; \mathbf{Z})$ act on the two dimensional torus T^2 in the obvious way.

Proposition 6.6. *Let E be the total space of the T^2-bundle over M_0 associated with the principal Γ_0-bundle $X_0 \to M_0 = X_0/\Gamma_0$. Then, for $j = 1, 2$, the manifolds $e_j/\Gamma \cap P_j$ are diffeomorphic to the manifold E.*

Proof. Let $X(P_0)$ be the set defined by Borel-Serre in [4]. Then

$$X(P_0) = X \times_A \bar{A}$$

where A is the subgroup of positive diagonal matrices in $SL(3; \mathbf{R})$,

$$\bar{A} = \{(x, y) \in \mathbf{R}^2 \mid x \geqq 0, y \geqq 0\},$$

A acts on X via the geodesic action, and on $\bar{A}$ using the roots of $SL(2; \mathbf{R})$. (See [3] or [4].) It is not difficult to see that $e(P_1) \subset X(P_0)$ is the set

$$e(P_1) = X \times_{A'} \bar{A}$$

where $A' = \{(x, y) \in \bar{A} \mid y = 0, x > 0\}$. It follows that $e(P_1) = X/A$ where

$$A_1 = \left\{ \begin{bmatrix} u^2 & & 0 \\ & 1/u & \\ 0 & & 1/u \end{bmatrix} \in A \right\}$$

acts on X via the geodesic action. If we identify X with P_0 in the usual way, then the geodesic action is given by $p \cdot a = a\, p$, $p \in P_0$, $a \in A$.

Let $H \subset SL(3; \mathbf{R})$ be defined by

$$H = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ & * & \end{bmatrix} \in SL(3; \mathbf{R}) \right\}$$

and $K \subset H$ by

$$K = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ 0 & & \\ 0 & & A \end{bmatrix} \in H,\ A \in SO(2) \right\}.$$

Now P_1 acts on H/K and on X/A_1, and it is easily seen that the inclusion $H \subset SL(3;\mathbf{R})$ defines a P_1-equivariant diffeomorphism of H/K onto X/A_1. If we define $f\colon H \to X_0 \times T^2$ by

$$f \begin{bmatrix} 1 & 0 & 0 \\ a & & \\ b & & M \end{bmatrix} = (M, (a, b))$$

then f defines a diffeomorphism of $\Gamma \cap P_1 \backslash H/K$ onto E.

In a similar fashion, one can prove

Proposition 6.7. *Let N be the subgroup of lower triangular matrices in $SL(3;\mathbf{R})$ with ones on the diagonal and $\Delta = \Gamma \cap N$ the subgroup of N. Then $e_0/\Gamma \cap P_0$ is diffeomorphic to N/Δ.*

If we define a manifold $\bar{E}$ to be the torus bundle over the complement of four *open* disks in S^2 (as in the definition of E), then $\bar{E}$ has four boundary components each diffeomorphic to N/Δ. In fact K_1 and K_2 are diffeomorphic to $\bar{E}$.

For example, one of the boundary components of $\bar{E}$ can be described as follows: Let E_1 be the torus bundle over a circle,

$$E_1 = T^2 \times_{\mathbf{Z}} \mathbf{R}^1$$

where the generator α of $\mathbf{Z}$ acts on $T^2 \times \mathbf{R}^1$ by

$$([x, y], z)\,\alpha = ([x + 3y, y], z + 3).$$

The mapping of N onto $T^2 \times \mathbf{R}^1$ given by

$$\begin{bmatrix} 1 & & \\ z & 1 & \\ x & y & 1 \end{bmatrix} \mapsto ([x, y], z)$$

defines a diffeomorphism of N/Δ onto E_1.

7. The $SL(3; F_3)$ Module $H_2(\Gamma(3;3))$

As a consequence of Proposition 6.5, we have a Mayer-Vietoris sequence

$$\cdots \to H_2 W_0 \xrightarrow{\gamma_2} H_2 W_1 \oplus H_2 W_2 \to H_2 \partial M \to H_1 W_0 \xrightarrow{\gamma_1} H_1 W_1 \oplus H_1 W_2 \to \cdots \tag{7.1}$$

where $M = \Gamma \backslash \bar{X}$ and W_1, W_2 are each the disjoint union of 13 copies of the manifold $\bar{E}$ defined after Proposition 6.7. We also have an exact sequence

$$\cdots \to H_3(M, \partial M) \xrightarrow{\partial} H_2 \partial M \xrightarrow{j_*} H_2 M \to H_2(M, \partial M) \to \cdots$$

where $H_2 M \simeq H_2 \Gamma$. (Recall $\Gamma = \Gamma(3;3)$.)

Proposition 7.1. *The mapping* $j_*: H_2 \partial M \to H_2 M$ *is surjective.*

Proof. In light of the exact sequence

$$H_2 \partial M \xrightarrow{j_*} H_2 M \to H_2(M, \partial M) \xrightarrow{\partial} H_1 \partial M \to H_1 M$$

we see that the proposition will follow from the injectivity of the mapping $\partial: H_2(M; \partial M) \to H_1 \partial M$.

Now

$$H_2(M; \partial M) \simeq H^3 M \simeq H_0(M; St) \simeq \overline{St}$$

by Lefschetz duality, Borel-Serre duality, and Theorem 1.2. Furthermore, ∂M clearly retracts onto $\bar{\mathscr{T}}(3)$ so $H_1 \partial M$ has $H_1 \bar{\mathscr{T}}(3) \simeq \overline{St}$ as a direct summand. Since $H_1 M$ is finite by Theorem 1.1 and $\overline{St}$ is a finitely generated free abelian group, it follows that $\partial: H_2(M; \partial M) \to H_1 \partial M$ is injective.

Let $\mathbf{C}$, $\mathbf{K}$, and $\mathbf{L}$ be the $SL(3; \mathbf{F}_3)$ modules defined by

$\mathbf{C} = \operatorname{coker} \gamma_2: H_2 W_0 \to H_2 W_1 \oplus H_2 W_2$,

$\mathbf{K} = \ker \quad \gamma_1: H_1 W_0 \to H_1 W_1 \oplus H_1 W_2$,

$\mathbf{L} = \text{Image } \partial: H_3(M, \partial M) \to H_2 \partial M$.

Then we have exact sequences of $SL(3; \mathbf{F}_3)$-modules

$$\begin{array}{ccccccccc} & & & & 0 & & & & \\ & & & & \downarrow & & & & \\ & & & & \mathbf{C} & & & & \\ & & & & \downarrow{\scriptstyle i} & & & & \\ 0 & \longrightarrow & \mathbf{L} & \longrightarrow & \mathrm{H}_2 \partial M & \xrightarrow{j^*} & H_2 M & \longrightarrow & 0 \\ & & & & \downarrow & & & & \\ & & & & \mathbf{K} & & & & \\ & & & & \downarrow & & & & \\ & & & & 0 & & & & \end{array} \tag{7.2}$$

Recall that $\mathbf{A}$ and $\mathbf{B}$ are the $SL(3; \mathbf{F}_3)$ modules defined by induction from the unique one dimensional non-trivial integral representations of P_1 and P_2. (See Section 1.)

Proposition 7.2. *The* $SL(3; \mathbf{F}_3)$ *module* $\mathbf{C}$ *is isomorphic to* $\mathbf{A} \oplus \mathbf{B}$.

This result is proved in Section 8. We also have the following.

Proposition 7.3. *The composite*

$$\mathbf{L} \to H_2\,\partial M \to \mathbf{K}$$

is injective with cokernel $\mathbf{Z}_2$.

The proof of this result is given in Sections 9, 10, and 11. Combining this with diagram (7.2) we have the

Corollary. The composite

$$\mathbf{C} \to H_2\,\partial M \to H_2 M \simeq H_2\,\Gamma$$

is injective with cokernel $\mathbf{Z}_2$.

Theorem 1.4 is now a consequence of this corollary and Proposition 7.2. We now determine $H^2(\Gamma)$.

Consider the universal coefficient sequence

$$0 \to \mathrm{Ext}(H_1(\Gamma); \mathbf{Z}) \to H^2(\Gamma) \to \mathrm{Hom}(H_2(\Gamma); \mathbf{Z}) \to 0.$$

We prove in Section 12 that

$$\mathrm{Ext}(H_1(\Gamma); \mathbf{Z}) \simeq \mathscr{M}(3;3)/J$$

where J is the subgroup of $\mathscr{M}(3;3)$ consisting of scalar matrices. Thus the sequence above becomes

$$0 \to \mathscr{M}(3;3)/J \to H^2(\Gamma) \to \mathrm{Hom}(H_2 M, \mathbf{Z}) \to 0. \tag{7.3}$$

This determines the $SL(3;\mathbf{F}_3)$ module $H^2\,\Gamma$ up to an extension.

Proposition 7.4. *The mapping* $\partial\colon H_3(M, \partial M) \to H_2\,\partial M$ *is injective so we have*

$$\mathbf{L} \simeq H_3(M, \partial M) \simeq H^2 M \simeq H^2\,\Gamma.$$

In Section 12, we will use this result to prove that the sequence (7.3) does not split over $\mathbf{Z}[\overline{SL}]$, where $\overline{SL} = SL(3;\mathbf{F}_3)$.

Proof. Consider the diagram of universal coefficient sequences

$$\begin{array}{ccccccc}
0 \to & \mathrm{Ext}(H_1 M; \mathbf{Z}) & \longrightarrow & H^2 M & \longrightarrow & \mathrm{Hom}(H_2 M; \mathbf{Z}) & \to 0 \\
& \downarrow & & \downarrow & & \downarrow & \\
0 \to & \mathrm{Ext}(H_1\,\partial M; \mathbf{Z}) & \longrightarrow & H^2\,\partial M & \longrightarrow & \mathrm{Hom}(H_2\,\partial M; \mathbf{Z}) & \to 0
\end{array}$$

The mapping $\mathrm{Ext}(H_1 M; \mathbf{Z}) \to \mathrm{Ext}(H_1\,\partial M; \mathbf{Z})$ is an isomorphism since $H_1\,\partial M \to H_1 M$ is an isomorphism on torsion. The mapping $\mathrm{Hom}(H_2 M; \mathbf{Z}) \to \mathrm{Hom}(H_2\,\partial M; \mathbf{Z})$ is injective since $H_2\,\partial M \to H_2 M$ is surjective by Proposition 7.1. It follows that $H^2 M \to H^2\,\partial M$ is injective and Proposition 7.4 follows by duality.

8. The Proof of Proposition 7.2

It will be convenient in the remainder of the paper to write Λ for $\mathbf{Z}[\overline{SL}]$ and Λ_i for $\mathbf{Z}[\bar{P}_i]$, $i = 0, 1, 2$.

Recall that, for $j=0, 1, 2$,

$$W_j = K_j \times_{P_j} \overline{SL}$$

where K_0, K_1 and K_2 are $K(\pi, 1)$-manifolds with

$$\pi_1 K_0 \simeq \Gamma_0 = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ a & 1 & 0 \\ b & c & 1 \end{bmatrix} \in \Gamma \right\} = \Delta,$$

$$\pi_1 K_1 \simeq \Gamma_1 = \left\{ \begin{bmatrix} 1 & 0 & 0 \\ a & & \\ b & & M \end{bmatrix} \in \Gamma, M \in \Gamma(2;3) \right\},$$

$$\pi_1 K_2 \simeq \Gamma_2 = \left\{ \begin{bmatrix} & M & & 0 \\ & & & 0 \\ b & & c & 1 \end{bmatrix} \in \Gamma, M \in \Gamma(2;3) \right\}.$$

Lemma 8.1. *There are isomorphisms of $\overline{SL}$-modules*

$$H_q W_j \simeq H_q K_j \otimes_{\Lambda_j} \Lambda$$

for $j=0, 1, 2$. Furthermore, under this identification, if $u \oplus g \in H_q W_0$, then $\gamma_2(u \otimes g) = (j_{1*}(u) \otimes g, j_{2*}(u) \otimes g)$ where $j_1 \colon W_0 \subset W_1$, $j_2 \colon W_0 \subset W_2$, and $\gamma_2 \colon H_2 W_0 \to H_2 W_1 \oplus H_2 W_2$.

The proof is trivial.

We now compute the homology of the manifolds K_0, K_1, and K_2.

Lemma 8.2. *For $j=1, 2$, we have $H_0 K_j \simeq \mathbf{Z}$, $H_q K_j = 0$ for $q>3$, and*

$$H_1 K_j \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}_3 \oplus \mathbf{Z}_3,$$

$$H_2 K_j \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z},$$

$$H_3 K_j \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}.$$

Proof. Let $\{E^r_{pq}, d_r\}$ be the spectral sequence of the torus fibration $T^2 \subset E \to M$ defined at the end of Section 6 (here M is the complement of four closed disks in S^2.) Since E is diffeomorphic to the interior of K_1 and K_2, $H_q E \simeq H_q K_j$, $j=1, 2$.

Now $E^2_{pq} \simeq H_p(M; H_q T^2) = 0$ for $p>1$ or $q>2$. Furthermore, it is straight forward to show that $E^2_{01} \simeq \mathbf{Z}_3 \oplus \mathbf{Z}_3$, $E^2_{02} \simeq \mathbf{Z}$, $E^2_{10} \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$, and $E^2_{12} \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$. To compute E^2_{11}, we use the standard resolution of a free group to get an exact sequence

$$0 \to H_1(M; H_1 T^2) \to (H_1 T^2)^3 \to H_1 T^2 \to H_0(M; H_1 T^2) \to 0.$$

It follows that $H_1(M; H_1 T^2)$ is free abelian of rank 4 and the lemma follows.

Lemma 8.3. *The homology groups of K_0 are given by*

$$H_0 K_0 \simeq H_3 K_0 \simeq \mathbf{Z},$$

$$H_1 K_0 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}_3,$$

$$H_2 K_0 \simeq \mathbf{Z} \oplus \mathbf{Z}.$$

Proof. The isomorphism $H_1 K_0 \simeq \mathbf{Z}\oplus\mathbf{Z}\oplus\mathbf{Z}_3$ follows from the fact that the commutator subgroup of $\pi_1 K_0$ is generated by the matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 9 & 0 & 1 \end{bmatrix}.$$

The remainder of the lemma follows from the fact that K_0 is a circle bundle over a torus. We leave the details to the reader.

Define two homomorphisms $\mathbf{Z}\oplus\mathbf{Z}\to\pi_1 K_0$ by

$$(s,t)\mapsto\begin{bmatrix} 1 & & \\ 3s & 1 & \\ 3t & 0 & 1 \end{bmatrix} \quad (s,t)\mapsto\begin{bmatrix} 1 & & \\ 0 & 1 & \\ 3t & 3s & 1 \end{bmatrix}.$$

These homomorphisms define mappings $\alpha,\beta\colon T^2\to K_0$. We let $u,v\in H_1 T^2$ be the usual generators $u\,v$ the corresponding generator of $H_2 T^2$, and define $a,b\in H_2 K_0$ by

$$a=\alpha_*(u\,v), \qquad b=\beta_*(u\,v).$$

It is easy to see that a and b generate $H_2 K_0$.

Let $f_1=I, f_2, f_3, f_4$ be the coset representatives for $\bar{P}_0\backslash\bar{P}_1$ given by

$$f_2=\begin{bmatrix} 1 & & \\ & 0 & 1 \\ & -1 & 0 \end{bmatrix}, \quad f_3=\begin{bmatrix} 1 & & \\ & 1 & 1 \\ & & 1 \end{bmatrix}, \quad f_4=\begin{bmatrix} 1 & & \\ & 1 & -1 \\ & & 1 \end{bmatrix},$$

and let $j_*\colon H_2\partial K_1\to H_2 K_1$ be induced by the inclusion $j\colon \partial K_1\subset K_1$. We identify $H_2\partial K_1$ with $H_2 K_0\otimes_{\Lambda_0}\Lambda_1$ and $H_2 K_1$ with $H_2 K_1\otimes_{\Lambda_1}\Lambda_1$ in the obvious way where $\Lambda_i=\mathbf{Z}[\bar{P}_i]$, $i=0,1,2$.

Lemma 8.4. *Let $a_i=j_*(a\otimes f_i)$, $b_i=j_*(b\otimes f_i)$ for $1\leqq i\leqq 4$. Then $a_i=a_1$ for $i=2,3,4$ and the elements a_1, b_1, b_2, b_3, b_4 form a $\mathbf{Z}$-base for $H_2 K_1$.*

Proof. Let $k_*\colon H_2 K_0\to H_2 K_1$ be induced by the inclusion $k\colon K_0\subset K_1$ (corresponding to the inclusion $\Gamma_0\subset\Gamma_1$). Then

$$a_i=j_*(a\otimes f_i)=k_*(a)\otimes f_i=k_*(a)f_i\otimes 1$$

where f_i acting on $H_2 K_1$ is induced by the inner automorphism $x\mapsto f_i^{-1}\times f_i$ of $\pi_1 K_1$ (considering f_i as an element of $SL(3;\mathbf{Z})$). The fact that $a_i=a_1$ for $i=2,3,4$ now follows by direct computation. For example,

$$f_2^{-1}\begin{bmatrix} 1 & & \\ 3s & 1 & \\ 3t & & 1 \end{bmatrix} f_2=\begin{bmatrix} 1 & & \\ -3t & 1 & \\ 3s & & 1 \end{bmatrix}$$

and it follows that

$$\begin{aligned} a_2 &= k_*\,\alpha_*((-v)\,u) \\ &= k_*\,\alpha_*(u\,v) \\ &= a_1. \end{aligned}$$

To prove that a_1, b_1, b_2, b_3, b_4 form a Z-base for $H_2 K_1$, we consider the exact sequence of the pair $(K_1, \partial K_1)$,

$$\cdots \to H_2 \partial K_1 \to H_2 K_1 \to H_2(K_1, \partial K_1) \to H_1 \partial K_1 \to H_1 K_1$$
$$\to H_1(K_1, \partial K_1) \to H_0 \partial K_1 \to H_0 K_1 \to 0.$$

Then $H_0 \partial K_1 \to H_0 K_1$ has kernel $\mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$ since ∂K_1 has four components, and

$$H_1(K_1, \partial K_1) \simeq H^3 K_1 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$$

by duality, the universal coefficient theorem, and Lemma 8.2. Thus $H_1(K_1, \partial K_1) \to H_0 \partial K_1$ is injective.

Similarly,

$$H_2(K_1, \partial K_1) \simeq H^2 K_1 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}_3 \oplus \mathbf{Z}_3$$

and it follows that $H_2(K_1, \partial K_1) \to H_1 \partial K_1$ is injective so $H_2 \partial K_1 \to H_2 K_1$ is surjective. Since $a \otimes f_i$, $b \otimes f_i$, $1 \leqq i \leqq 4$, generate $H_2 \partial K_1$, a_1, b_1, b_2, b_3, b_4 must generate $H_2 K_1$. But $H_2 K_1 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$ by Lemma 8.2 so these elements must form a $\mathbf{Z}$-base for $H_2 K_1$.

This completes the proof of Lemma 8.4.

Let i_*: $H_2 \partial K_2 \to H_2 K_2$ be induced by the inclusion i: $\partial K_2 \subset K_2$. We identify $H_2 \partial K_2$ with $H_2 K_0 \otimes_{\Lambda_0} \Lambda_2$ and $H_2 K_2$ with $H_2 K_2 \otimes_{\Lambda_2} \Lambda_2$ in the obvious way.

Lemma 8.5. *Let $c_j = i_*(a \otimes f_j)$, $d_j = i_*(b \otimes f_j)$ for $1 \leqq j \leqq 4$. Then $d_j = d_1$ for $j = 2, 3, 4$ and the elements c_1, c_2, c_3, c_4, d_1 form a $\mathbf{Z}$-base for $H_2 K_2$.*

The proof is analogous to the proof of Lemma 8.4 and we omit it.

Note that $\bar{P}_1$ and $\bar{P}_2$ are subgroups of $\overline{SL}$ consisting of matrices of the form

$$\begin{bmatrix} \pm & 1 & 0 & 0 \\ r & & & \\ s & & M & \end{bmatrix}, \quad \begin{bmatrix} & & & 0 \\ & M & & 0 \\ s & t & & \pm 1 \end{bmatrix},$$

respectively. Let $\bar{P}_1^0$ and $\bar{P}_2^0$ be the subgroups defined by

$$\bar{P}_1^0 = \{(a_{ij}) \in \bar{P}_1 \mid a_{11} = 1\},$$
$$\bar{P}_2^0 = \{(a_{ij}) \in \bar{P}_2 \mid a_{33} = 1\}.$$

Lemma 8.6. *The subgroup of $H_2 K_1$ fixed under the action of $\bar{P}_1^0$ is generated by a_1 and the subgroup of $H_2 K_2$ fixed under the action of $\bar{P}_2^0$ is generated by d_1. Furthermore, for $p_j \in \bar{P}_j$, $p_j \notin \bar{P}_j^0$, we have $a_1 p_1 = -a_1$ and $d_1 p_2 = -d_2$.*

Proof. The fact that a_1 is fixed under $\bar{P}_1^0$ and d_1 is fixed under $\bar{P}_2^0$ is a straightforward computation similar to what was done in the proof of Lemma 8.4. To see that a_1 (respectively d_1) generates the subgroup fixed under the action of $\bar{P}_1^0$ (respectively $\bar{P}_2^0$), we set

$$p = \begin{bmatrix} 1 & & \\ & -1 & \\ & & -1 \end{bmatrix}, \quad q = \begin{bmatrix} -1 & & \\ & -1 & \\ & & 1 \end{bmatrix},$$

and compute to show that $b_j p = -b_j$, $c_j q = -c_j$ for $1 \leqq j \leqq 4$.

Finally, if $p_j \in \bar{P}_j$, $p_j \notin \bar{P}_j^0 \cdot j = 1, 2$, are defined by

$$p_1 = p_2 = \begin{bmatrix} -1 & & \\ & 1 & \\ & & -1 \end{bmatrix},$$

then $a_1 p_1 = -a_1$, $d_1 p_2 = -d_2$ by direct computation.

Let $g_i \in \overline{SL}$, $1 \leqq i \leqq 13$, be coset representatives for $\overline{SL}/\bar{P}_1$. It is easy to see that the g_i can be chosen so as to also be coset representatives for $\overline{SL}/\bar{P}_2$. Define elements $u_1 \in H_2 W_1$, $v_i \in H_2 W_2$ by $u_i = a_1 \otimes g_i$, $v_i = d_1 \otimes g_i$, $1 \leqq i \leqq 13$.

Lemma 8.7. *The elements u_i, v_i, $1 \leqq i \leqq 13$, form a* **Z***-base for* $\mathbf{C} = coker\ \gamma_2 : H_2 W_0 \to H_2 W_1 \oplus H_2 W_2$.

Proof. By the results above, the elements $a \otimes f_j g_i$, $b \otimes f_j g_i$, $1 \leqq i \leqq 13$, $1 \leqq j \leqq 4$, form a **Z**-base for $H_2 W_0$ and the elements $u_i, v_i, u_i + c_j g_i$, $b_j g_i + v_i$, $1 \leqq i \leqq 13$, $1 \leqq j \leqq 4$, form a **Z**-base for $H_2 W_1 \oplus H_2 W_2$. Since

$$\gamma_2(a \otimes f_j g_i) = u_i + c_j g_i,$$

$$\gamma_2(b \otimes f_j g_i) = b_j g_i + v_i,$$

the lemma is proved.

To complete the proof of Proposition 7.2, we note that, for any $g \in \overline{SL}$,

$$\begin{aligned} u_i g &= u_1 g_i g \\ &= u_1 p g_k \quad \text{for some } k = 1, \ldots, 13,\ p \in \bar{P}_1 \\ &= \begin{cases} u_k & \text{if } p \in \bar{P}_1^0, \\ -u_k & \text{if } p \notin \bar{P}_1^0. \end{cases} \end{aligned}$$

Similarly, $v_i g = \pm v_k$ for $g \in \overline{SL}$. It now follows from the definition of induced representation that $\mathbf{C} \simeq \mathbf{A} \oplus \mathbf{B}$.

9. The Kernel of γ_1

Before proving Proposition 7.3, we determine, modulo torsion, the kernel **K** of the homomorphism $\gamma_1 : H_1 W_0 \to H_1 W_1 \oplus H_1 W_2$ which occurs in the Mayer-Vietoris sequence (7.1).

Recall that $\mathcal{M}(3; 3)$ is the additive group of 3×3 matrices with entries in $\mathbf{F}_3$. Let S be the subset of $\mathcal{M}(3; 3)$ consisting of all matrices e with $e^2 = 0$ and let U_0 be the free abelian group generated by S with the relation $n(a_{ij}) = (n a_{ij})$ for any integer n, $(a_{ij}) \in S$. In particular, U_0 is an elementary abelian 3-group. Note that $\overline{SL}$ acts on U_0 on the right by conjugation.

Lemma 9.1. *There is an $\overline{SL}$-isomorphism of U_0 onto the torsion submodule of $H_1 W_0$.*

Proof. It follows easily from Lemmas 8.1 and 8.3 that the torsion submodule of $H_1 W_0$ can be identified with the direct sum of the centers of the minimal parabolic subgroups of $SL(3; \mathbf{F}_3)$. If g is a central element of a minimal parabolic subgroup

of $SL(3;\mathbf{F}_3)$, then $(g-I)^2=0$ so $(g-I)\in S$. (Since all minimal parabolic subgroups of $SL(3;\mathbf{F}_3)$ are conjugate, this need only be checked for one minimal parabolic subgroup.) If $e\in S$, then $I+e\in SL(3;\mathbf{F}_3)$ and the centralizer of $I+e$ is a minimal parabolic subgroup (Since any two elements e, e' in S, $e\neq -e'$, are conjugate, this need only be checked for one e in S.) The lemma now follows.

For any element $\lambda\in\mathcal{M}(3;3)$, we define $\ker\lambda$ and $\operatorname{Im}\lambda$ to be the usual kernel and image of λ considered as a linear transformation acting on the right on $\mathbf{F}_3^3$.

Let U_1 be the free abelian group generated by S with relations

(i) $n(a_{ij})=(n\,a_{ij})$,

(ii) if $\alpha=(a_{ij})$, $\beta=(b_{ij})\in S$ with $\operatorname{Im}\alpha=\operatorname{Im}\beta$, then $\alpha+\beta=(a_{ij}+b_{ij})$.

Similarly, we define U_2 to be the free abelian group generated by S with relation (i) above and

(iii) if $\alpha=(a_{ij})$, $\beta=(b_{ij})\in S$ with $\ker\alpha=\ker\beta$, then $\alpha+\beta=(a_{ij}+b_{ij})$.

Then U_1 and U_2 are elementary abelian 3-groups, $\overline{SL}$ acts on both U_1 and U_2, and we let $\rho\colon U_0\to U_1\oplus U_2$ be the obvious $\overline{SL}$-homomorphism.

Lemma 9.2. *There is an $\overline{SL}$-isomorphism of $U_1\oplus U_2$ onto the torsion submodule of $H_1W_1\oplus H_1W_2$ such that the diagram*

$$\begin{array}{ccc} H_1W_0 & \xrightarrow{\gamma_2} & H_1W_1\oplus H_1W_2 \\ \uparrow & & \uparrow \\ U_0 & \xrightarrow{\rho} & U_1\oplus U_2 \end{array}$$

is commutative.

Proof. Recall that H_1W_1 is the direct sum of 13 copies of H_1K_1, one for each parabolic subgroup of $\overline{SL}$ conjugate to $\bar{P}_1$. Let e be a generator of U_1, $P(e)$ the minimal parabolic subgroup of $\overline{SL}$ which is the centralizer of $I+e$, and P the parabolic subgroup of $\overline{SL}$ conjugate to P_1 and containing $P(e)$. Then $I+3e$ (as an element of $SL=SL(3;\mathbf{Z})$) determines an element in the torsion subgroup $\operatorname{tor}H_1W_1$. It is easy to check that this defines a $\mathbf{Z}[\overline{SL}]$-mapping of U_1 into $\operatorname{tor}H_1W_1$.

To see that this mapping is an isomorphism, we construct an inverse from $\operatorname{tor}H_1W_1$ onto U_1.

Let e_1, e_2 in S be given by

$$e_1=\begin{bmatrix}0&&\\1&&\\0&&\end{bmatrix},\qquad e_2=\begin{bmatrix}0&&\\0&&\\1&&\end{bmatrix}.$$

Then $\operatorname{tor}H_1K_1\simeq\mathbf{Z}_3\oplus\mathbf{Z}_3$ is generated by $I+3e_1, I+3e_2$. According to Lemma 8.1, $\operatorname{tor}H_1W_1$ is the $\mathbf{Z}[\overline{SL}]$-module obtained by induction from the Λ_1-module $\operatorname{tor}H_1K_1$. It follows that the mapping of $\operatorname{tor}H_1K_1$ into U_0 taking $I+3e_j$ onto e_j, $j=1,2$, extends to a unique $\mathbf{Z}[\overline{SL}]$ mapping of H_1W_1 into U_1. It is straightforeward to verify that this mapping is an inverse to the mapping of U_1 into $\operatorname{tor}H_1W_1$ defined above.

A similar argument proves that U_2 is isomorphic to tor $H_1 W_2$. This completes the proof of Lemma 9.2.

Let u be the element of $H_1 K_0$ defined by the matrix

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & 3 & 1 \end{bmatrix}.$$

and define $\sigma \in H_1 W_0$ by

$$\sigma = \sum_{j=1}^{4} u \otimes f_j$$

where the f_j, $1 \leqq j \leqq 4$, are the coset representatives for $\bar{P}_0 \backslash \bar{P}_1$ defined in section 8. Then, if $j_1 \colon W_0 \subset W_1, j_2 \colon W_0 \subset W_2$, we have

$$j_{1*}\sigma = (u + uf_1 + uf_2 + uf_3) \otimes 1$$

and $u + uf_1 + uf_2 + uf_3$ is represented by the matrix

$$(f_4^{-1} u f_4)(f_2^{-1} u f_2)(f_3^{-1} u f_3) u$$

which is easily seen to be the identify. Thus $j_{1*}\sigma = 0$. It also follows easily that $j_{2*}\sigma$ is the torsion element in $H_1 W_2$ which corresponds to the element

$$\begin{bmatrix} & & \\ 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & \\ -1 & \\ 0 & \end{bmatrix} + \begin{bmatrix} -1 & -1 \\ 1 & 1 \end{bmatrix} + \begin{bmatrix} 1 & -1 \\ 1 & -1 \end{bmatrix}$$

in U_2.

Let α, β, and x be the elements in tor $H_1 W_0$ defined by

$$\alpha = \begin{bmatrix} 0 & \\ 1 & \\ 0 & \end{bmatrix} + \begin{bmatrix} 0 & \\ 1 & \\ 1 & \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix},$$

$$\beta = \begin{bmatrix} 0 & \\ 0 & \\ -1 & \end{bmatrix} + \begin{bmatrix} 0 & \\ -1 & \\ -1 & \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix},$$

$$x = \alpha \begin{bmatrix} & & 1 \\ & 1 & \\ -1 & & \end{bmatrix} - \alpha \begin{bmatrix} 1 & & \\ & & 1 \\ 1 & & \end{bmatrix} + (\alpha + \beta) \begin{bmatrix} 1 & 1 & 1 \\ & 1 & \\ & & 1 \end{bmatrix}$$

$$-(\alpha + \beta) \begin{bmatrix} 1 & -1 & -1 \\ & 1 & \\ & & 1 \end{bmatrix} + (\alpha - \beta) \begin{bmatrix} 1 & -1 & 1 \\ & 1 & \\ & & 1 \end{bmatrix} - (\alpha - \beta) \begin{bmatrix} 1 & 1 & -1 \\ & 1 & \\ & & 1 \end{bmatrix}.$$

Here $\alpha g = g^{-1} \alpha g$ for $g \in \bar{SL}$.

Lemma 9.3. *The element $\sigma + x$ in $H_1 W_0$ is in the kernel of*

$$\gamma_1 \colon H_1 W_0 \to H_1 W_1 \oplus H_1 W_2 .$$

Proof. Note that $j_{1*}\alpha = j_{1*}\beta = 0$ by the defining relations in U_0. It follows that $j_{1*}x = 0$ so, since $j_{1*}\sigma = 0$, we have $j_{1*}(\sigma + x) = 0$.

To complete the proof of Lemma 9.3, we need to observe that $j_{1*}x = -j_{1*}\sigma$. This is a lengthy but straightforward computation using the defining relations for U_2. We leave it as an exercise for the reader.

Similarly, we let v be the element of $H_1 K_0$ defined by the matrix

$$\begin{bmatrix} 1 & & \\ 3 & 1 & \\ & & 1 \end{bmatrix}$$

and $\tau \in H_1 W_0$ by

$$\tau = \sum_{j=1}^{4} v \otimes f_j'$$

where f_j' are the coset representatives for $\bar{P}_0 \backslash \bar{P}_2$ given by

$$f_1' = I, \quad f_2' = \begin{bmatrix} 0 & 1 & \\ -1 & 0 & \\ & & 1 \end{bmatrix}, \quad f_3' = \begin{bmatrix} 1 & 1 & \\ & 1 & \\ & & 1 \end{bmatrix}, \quad f_4' = \begin{bmatrix} 1 & -1 & \\ & 1 & \\ & & 1 \end{bmatrix}.$$

Then just as above, we can define α', β', and x' in $\operatorname{tor} H_1 W_1$ such the $\gamma_1(\tau + x') = 0$. We omit the details.

Lemma 9.4. *Let* $\operatorname{tor}\mathbf{K}$ *be the torsion submodule of* $\mathbf{K} = \ker\gamma_1$. *Let* $\tilde{\sigma}$, $\tilde{\tau}$ *be the images in* $\mathbf{K}/\operatorname{tor}\mathbf{K}$ *of* $\sigma + x$, $\tau + y$ *and* $\tilde{\mathbf{A}}$, $\tilde{\mathbf{B}}$ *the* $\overline{SL}$*-modules of* $\mathbf{K}$ *generated by* $\tilde{\sigma}$, $\tilde{\tau}$ *respectively. Then* $\mathbf{K}/\operatorname{tor}\mathbf{K} \simeq \tilde{\mathbf{A}} \oplus \tilde{\mathbf{B}}$, $\tilde{\mathbf{A}} \simeq \mathbf{A}$, *and* $\tilde{\mathbf{B}} \simeq \mathbf{B}$ *where* $\mathbf{A}$ *and* $\mathbf{B}$ *are defined in Section 1.*

Proof. Let T_0, T_1, T_2 be the torsion submodules of $H_1 K_0$, $H_1 K_1$, $H_1 K_2$ respectively. It is easy to see that the following isomorphisms exist.

$$H_1 W_0 / \operatorname{tor} H_1 W_0 \simeq (H_1 k_0 / T_0) \otimes_{\Lambda_0} \Lambda,$$
$$H_1 W_1 / \operatorname{tor} H_1 W_1 \simeq (H_1 K_1 / T_1) \otimes_{\Lambda_1} \Lambda,$$
$$H_1 W_2 / \operatorname{tor} H_1 W_2 \simeq (H_1 K_2 / T_2) \otimes_{\Lambda_2} \Lambda.$$

(Recall $\Lambda = \mathbf{Z}[\overline{SL}]$ and $\Lambda_j = \mathbf{Z}[\bar{P}_j]$, $j = 0, 1, 2$.) If $\tilde{u}, \tilde{v}$ are the elements of $H_1 K_0 / T_0$ determined by $u, v \in H_1 K_0$, then $H_1 K_0 / T_0$ is the sum of the two Λ_0-submodules H, H' generated by $\tilde{u}, \tilde{v}$ respectively. It follows that

$$H_1 W_0 / \operatorname{tor} H_1 W_0 \simeq H \otimes_{\Lambda_0} \Lambda + H' \otimes_{\Lambda_0} \Lambda .$$

Now $j_{2*}\tilde{u} = 0 = j_{1*}\tilde{v}$ where $j_1 \colon W_0 \subset W_1$ and $j_2 \colon W_0 \subset W_2$ so

$$\ker\tilde{\gamma}_1 = \ker j_{1*} \oplus \ker j_{2*}$$

where

$$\tilde{\gamma}_1 \colon H_1 W_0/\operatorname{tor} H_1 W_0 \to H_1 W_1/\operatorname{tor} H_1 W_1 \oplus H_1 W_2/\operatorname{tor} H_1 W_2$$

is induced by γ_1.

If we write

$$H \otimes_{\Lambda_0} \Lambda \simeq (H \otimes_{\Lambda_0} \Lambda_1) \otimes_{\Lambda_1} \Lambda,$$

we can identify

$$j_{1*} \colon H_1 W_0/\operatorname{tor} H_1 W_0 \to H_1 W_1/\operatorname{tor} H_1 W_1$$

with the mapping

$$h_* \otimes 1 \colon (H \otimes_{\Lambda_0} \Lambda_1) \otimes_{\Lambda_1} \Lambda \to (H_1 K_1/T_1) \otimes_{\Lambda_1} \Lambda$$

where h_* is the composite

$$H \otimes_{\Lambda_0} \Lambda_1 \subset (H_1 K_0/T_0) \otimes_{\Lambda_0} \Lambda_1 \to H_1 K_1/T_1.$$

Since Λ is Λ_1-free, it follows that

$$\ker j_{1*} \simeq (\ker h_*) \otimes_{\Lambda_1} \Lambda.$$

By definition,

$$H \otimes_{\Lambda_0} \Lambda_1 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}$$

and, by Lemma 8.2,

$$H_1 K_1/T_1 \simeq \mathbf{Z} \oplus \mathbf{Z} \oplus \mathbf{Z}.$$

It follows easily now that $\tilde{\sigma}$ generates $\ker h_*$.

The argument for $\ker j_{2*}$ is similar and we omit it.

To complete the proof of Lemma 9.4, we need only show that $\sigma t = -\sigma$ for $t \in \bar{P}_1$, $t \notin \bar{P}_1^0$. It is enough to check this for

$$t = \begin{bmatrix} -1 & & \\ & -1 & \\ & & 1 \end{bmatrix}.$$

Now $\sigma = u \otimes 1 + u \otimes f_2 + u \otimes f_3 + u \otimes f_4$ where $u \in H_1 K_0$ is defined by the matrix

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & 3 & 1 \end{bmatrix}$$

and direct computation shows that

$$(u \otimes 1)t = u \otimes t = ut \otimes 1$$

where ut is defined by

$$t^{-1}\begin{bmatrix} 1 & & \\ & 1 & \\ & 3 & 1 \end{bmatrix} t = \begin{bmatrix} 1 & & \\ & 1 & \\ & -3 & 1 \end{bmatrix}.$$

Thus $(u\otimes 1)t = -u\otimes 1$. In the same way, we can show that

$$u\otimes f_2 t = u\otimes sf_2$$
$$= -u\otimes f_2,$$
$$u\otimes f_3 t = -u\otimes f_4,$$
$$u\otimes f_4 t = -u\otimes f_3,$$

where

$$s = \begin{bmatrix} -1 & & \\ & 1 & \\ & & -1 \end{bmatrix}.$$

This completes the proof of Lemma 9.4.

10. The Proof of Proposition 7.3

In this section, we prove the first half of Proposition 7.3, namely that the composite $\mathbf{L}\to H_2\partial M\to \mathbf{K}$ is injective. The remainder of the proof of Proposition 7.3 will be given in section 11.

We begin with the following.

Proposition 10.1. *There is an element z in $H_2\partial M$ with $j_* z = 0$ and such that the composite*

$$H_2\partial M\to \mathbf{K}\to \mathbf{K}/\mathrm{tor}\,\mathbf{K}$$

takes z onto the element

$$\tilde{z} = \tilde{\sigma} - \tilde{\tau}\begin{bmatrix} & & 1 \\ & 1 & \\ -1 & & \end{bmatrix}$$

where $\tilde{\sigma}$ and τ are defined in Lemma 9.4.

Before proving this proposition, we derive from it the fact that $\mathbf{L}\to H_2\partial M\to \mathbf{K}$ is injective.

Let

$$\varepsilon\colon\ \mathbf{K}/\mathrm{tor}\,\mathbf{K}\simeq \mathbf{A}\oplus\mathbf{B}\to \mathbf{Z}_2$$

be the augmentation mapping defined by $\varepsilon(\tilde{\sigma})=\varepsilon(\tilde{\tau})=1$ and let $\tilde{z}\in\mathbf{K}/\mathrm{tor}\,\mathbf{K}$ as above.

Lemma 10.2. *The kernel of the mapping* $\varepsilon\colon \mathbf{K}/\mathrm{tor}\,\mathbf{K} \to \mathbf{Z}_2$ *is generated as an* Λ*-module by* $\tilde{z}$.

Proof. First we observe that, as an abelian group $\ker\varepsilon$ is generated by elements of the form

$$2\tilde{\sigma}g, \quad 2\tilde{\tau}g, \quad \tilde{\sigma}-\tilde{\sigma}g, \quad \tau-\tilde{\tau}g, \quad \tilde{\sigma}-\tilde{\tau}$$

where $g\in\overline{SL}$. In fact, as an Λ-module, $\ker\varepsilon$ is generated by $2\tilde{\sigma}$, $2\tilde{\tau}$, and $\tilde{\sigma}-\tilde{\tau}$. To see this, let D be the Λ-submodule of $\ker\varepsilon$ generated by $2\tilde{\sigma}$, $2\tilde{\tau}$, and $\tilde{\sigma}-\tilde{\tau}$. Clearly $2\tilde{\sigma}g$, $2\tilde{\tau}g$, and $\tilde{\sigma}-\tilde{\tau}\in D$ and, if we can show that $\tilde{\sigma}-\tilde{\sigma}g$, $\tilde{\tau}-\tilde{\tau}g$ are in D, we will have shown that $D=\ker\varepsilon$.

Suppose $g_1\in\bar{P}_1$, $g_2\in\bar{P}_2$. Then

$$\tilde{\sigma}-\tilde{\sigma}g_1=0 \quad \text{or} \quad 2\tilde{\sigma}.$$

$$\tilde{\tau}-\tilde{\tau}g_2=0 \quad \text{or} \quad 2\tilde{\tau},$$

so $\tilde{\sigma}-\tilde{\sigma}g_1$, $\tilde{\tau}-\tilde{\tau}g_2\in D$. We also have

$$\tilde{\sigma}-\tilde{\sigma}g_2=(\tilde{\sigma}-\tilde{\tau})-(\tilde{\sigma}-\tilde{\tau})g_2-(\tilde{\tau}-\tilde{\tau}g_2),$$

$$\tilde{\sigma}-\tilde{\sigma}g_1g_2=(\tilde{\sigma}-\tilde{\sigma}g_2)+(\tilde{\sigma}-\tilde{\sigma}g_1)g_2$$

so $\tilde{\sigma}-\tilde{\sigma}g_1g_2\in D$. Now, $\overline{SL}$ is generated by $\bar{P}_1$ and $\bar{P}_2$ so we conclude that $\tilde{\sigma}-\tilde{\sigma}g\in D$. A similar argument holds for the element $\tilde{\tau}-\tilde{\tau}g$.

To complete the proof of Lemma 10.2, we need only show that $2\tilde{\sigma}$, $2\tilde{\tau}$, and $\tilde{\sigma}-\tilde{\tau}$ are in the Λ-submodule of $\ker\varepsilon$ generated by $\tilde{z}$.

Define $\tilde{\rho}$ by

$$\tilde{\rho}=\tilde{\tau}\begin{bmatrix} & & 1\\ & 1 & \\ -1 & & \end{bmatrix}.$$

Then

$$\begin{aligned}\tilde{\sigma}-\tilde{\tau}&=(\tilde{\sigma}-\tilde{\rho})+(\tilde{\rho}-\tilde{\tau})\\ &=\tilde{z}+\left(\tilde{\rho}-\tilde{\rho}\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}\begin{bmatrix}1 & & \\ & 1 & \\ 1 & & 1\end{bmatrix}\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}\right)\\ &=\tilde{z}+\left(\tilde{\rho}-\tilde{\rho}\begin{bmatrix}1 & & \\ & 1 & \\ 1 & & 1\end{bmatrix}\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}\right)\end{aligned}$$

since

$$\tilde{\rho}\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}=\tilde{\rho}.$$

Now $\tilde{\rho}=\tilde{\sigma}-\tilde{z}$ so

$$\tilde{\sigma}-\tilde{\tau}=\tilde{z}-\left(\tilde{z}-\tilde{z}\begin{bmatrix}1 & & \\ & 1 & \\ 1 & & 1\end{bmatrix}\right)\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}.$$

On the other hand, we have

$$\begin{aligned}2\tilde{\sigma}&=(\tilde{\sigma}-\tilde{\tau})-(\tilde{\sigma}-\tilde{\tau})\begin{bmatrix}-1 & & \\ & -1 & \\ & & 1\end{bmatrix}\\ &=2\tilde{z}-\tilde{z}\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}+\tilde{z}\begin{bmatrix}1 & -1 & \\ & 1 & \\ 1 & & \end{bmatrix}+\tilde{z}\begin{bmatrix}-1 & & 1\\ & -1 & \\ & & 1\end{bmatrix}-\tilde{z}\begin{bmatrix}-1 & & -1\\ & -1 & \\ -1 & & \end{bmatrix}\end{aligned} \tag{10.1}$$

A similar equation holds for $2\tilde{\tau}$ so Lemma 10.2 is proved.

Lemma 10.3. *The group H_2M is a torsion free group of rank 26.*

Proof. The fact that H_2M is torsion free follows from the fact that H^3M is torsion free (by the corollary to Theorem 1.2). Furthermore the Euler characteristic of M is zero (see Harder [5] or Borel-Serre [4], page 182) and the fact that rank $H_2M=26$ follows from Theorem 1.1 and the corollary to Theorem 1.2 above.

We can now prove the first half of Proposition 7.3.

Lemma 10.4. *The composite*

$$\mathbf{L}\to H_2\partial M\to\mathbf{K}$$

is injective.

Proof. Since H_2M and $\mathbf{C}$ are torsion free, it follows from the diagram in Section 7 that the composite above is injective on torsion. Furthermore, since rank $H_2\partial M=52$ and rank $H_2M=26$, it follows that rank $\mathbf{L}=26$. Now, according to Lemma 10.2, the composite

$$\mathbf{L}\to H_2\partial M\to\mathbf{K}\to\mathbf{K}/\mathrm{tor}\,\mathbf{K}$$

has as image a free abelian group of rank 26. It follows that $\mathbf{L}/\mathrm{tor}\,\mathbf{L}\to\mathbf{K}/\mathrm{tor}\,\mathbf{K}$ is injective and the proof of Lemma 10.4 is complete.

We now turn to the proof of Lemma 10.1.

Recall that

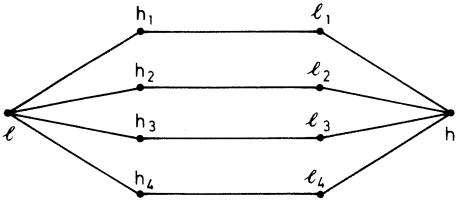
$$\partial M=W_1\underset{W_0}{\cup}W_2$$

where

$$W_i=K_i\times_{\overline{P}_i}\overline{SL}$$

for $i=0,1,2$ and each of the K_i are $K(\pi,1)$ manifolds with fundamental groups isomorphic to subgroups of Γ (See the beginning of section 8).

Let l, l_1, l_2, l_3, l_4 be the line in Q^3 containing the points $(1,0,0)$, $(0\,1\,0)$, $(0\,0\,1)$, $(0\,1\,-1)$, and $(0\,1\,1)$ respectively. Let h be the plane in $\mathbf{Q}^3$ containing the lines $l_1, \ldots, l_4$ and h_i the plane in $\mathbf{Q}^3$ containing the lines l and l_i, $i=1,2,3,4$. It is easy to see that these lines and planes define a subcomplex S of the Tits building $\mathscr{T}(3)$ for $\mathbf{Q}^3$.



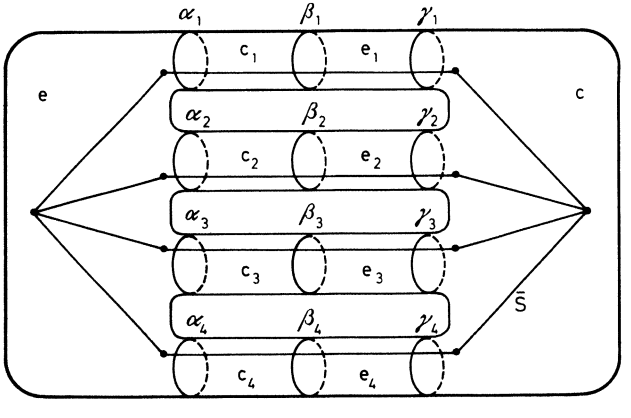
(10.2)

In fact, these lines and planes define, in the obvious way, a subcomplex $\bar{S}$ homeomorphic to S of the Tits building $\bar{\mathscr{T}}(3)$ for $\mathbf{F}_3^3$. The subcomplex $\bar{S}$ defines a submanifold N of ∂M and there is an embedding of S into the submanifold $\tilde{N}$ of $\bar{X}$ over N which is a homotopy equivalence. In addition, this embedding can be chosen so that the restriction to S of the mapping $\bar{X} \to \bar{X}/\Gamma$ is a homeomorphism onto $\bar{S} \subset N$.

Now N is the union of five copies of K_1, one for each of the lines $l, l_1, \ldots, l_4$ together with five copies of K_2, one for each of the planes $h, h_1, \ldots, h_4$. These submanifolds are identified along boundary components each diffeomorphic to K_0. We will, for the remainder of this section, use the notation $K(l)$, $K(l_i)$, $K(h)$, $K(h_i)$, $K(l, h_i)$, $K(l_i, h_i)$, $K(l_i, h)$, $i=l, \ldots, 4$, for these submanifolds of N. In particular, $K(l)$ and $K(l_i)$ are diffeomorphic to K_1, $K(h)$ and $K(h_i)$ are diffeomorphic to K_2, and $K(l, h_i)$, $K(l_i, h_i)$ and $K(l_i, h)$ are diffeomorphic to K_0.

We pick a basepoint in each of these submanifolds on the 1-complex $\bar{S}$ and use the edges of $\bar{S}$ to compare the fundamental groups of these submanifolds. In this way, we can identify these fundamental groups with subgroups if Γ.

Let c and e be copies of the complement of four open disks in the 2-sphere S^2 with $\partial e = \alpha_1 \cup \alpha_2 \cup \alpha_3 \cup \alpha_4$ and $\partial c = \gamma_1 \cup \gamma_2 \cup \gamma_3 \cup \gamma_4$. Let $c_1, \ldots, c_4$, $e_1, \ldots, e_4$ be eight copies of the cylinder $S^1 \times I$ with $\partial c_i = \alpha_i' \cup \beta_i$, $\partial e_i = \beta_i' \cup \gamma_i'$, $i=1, \ldots, 4$. Identifying α_i with α_i', β_i with β_i', and γ_i with γ_i', we obtain the closed surface F of genas three shown below



We have also shown an embedding of $\bar{S}$. We now describe a mapping φ of F into $N \subset \partial M$.

To begin with, φ takes $\bar{S} \subset F$ homeomorphically onto $\bar{S} \subset N$. Next φ will take each of the circles β_i into $K(l_1, h_i)$ so as to define the following elements of $\pi_1(K(l_i, h_i), *)$, $i = 1, \ldots, 4$. (The basepoints are always those chosen above on $\bar{S}$.)

$$\begin{bmatrix} 1 & & \\ & 1 & \\ -3 & & 1 \end{bmatrix}, \quad \begin{bmatrix} 1 & & \\ & -2 & 3 \\ & -3 & 4 \end{bmatrix}, \quad \begin{bmatrix} 1 & & \\ & 1 & 3 \\ & & 3 \end{bmatrix}, \quad \begin{bmatrix} 1 & & \\ & 4 & 3 \\ & -3 & -2 \end{bmatrix}.$$

In fact, these matrices also define elements of $\pi_1(K(l, h_i), *)$ and it is easy to see that φ can be defined over the cylinders c_i taking c_i into $K(h_i)$, α_i into $K(l, h_i)$ so that $\varphi | \alpha_i$ define the elements of $\pi_1(K(l, h_i), *)$ described above. Since the product in the order given of the four matrices above is the identity, we can extend φ to e.

In a similar way, we can extend φ over the cylinders e_i taking e_i into $K(l_i)$, γ_i into $K(l_i, h)$ so as to define the elements of $\pi_1(K(l_i, h), *)$ determined by the matrices above. As above φ extends to c because the product of the matrices above is the identity.

Lemma 10.5. *Let $[F] \in H_2 F$ be the fundamental class of F and let $z \in H_2 \partial M$ be the image of $[F]$ under the composite*

$$H_2 F \xrightarrow{\varphi^*} H_2 N \to H_2 \partial M.$$

Then z satisfies the conditions of Proposition 10.1.

Proof. Note first that the composite

$$\bar{S} \xrightarrow{\varphi | \bar{S}} N \subset \partial M \subset M$$

factors through $\bar{X}$ (by the definition of $\bar{S}$) so is null homotopic. It follows easily that the homomorphism $\pi_1 F \to \pi_1 M$ factors through a free group Σ on three generators. Since F and M are both $K(\pi; 1)$ spaces, the mapping $F \to M$ is determined by the homomorphism $\pi_1 F \to \pi_1 M$ so we can factor it through $K(\Sigma; 1)$. However, $H_2 K(\Sigma; 1) = 0$ so the $j_* \varphi_* [F] = j_* z = 0$.

Consider the decomposition of $F = F_1 \cup F_2$ where $F_1 = e \cup e_1 \cup \cdots \cup e_4$, $F_2 = c \cup c_1 \cup \cdots \cup c_4$. The Mayer-Vietoris sequence for this decomposition maps into the Mayer-Vietoris sequence for $\partial M = W_1 \cup W_2$ and we have the commutative diagram

$$\begin{array}{ccc} H_2 \partial M & \longrightarrow & H_1 W_0 \\ \uparrow{\scriptstyle \varphi_*} & & \uparrow{\scriptstyle \varphi_*} \\ H_2 F & \xrightarrow{\partial} & \bigoplus_{i=1}^{4} (H_1 \alpha_i \oplus H_1 \beta_i \oplus H_1 \gamma_i). \end{array}$$

Since

$$\partial [F] = \sum_{i=1}^{4} ([\gamma_i] - [\alpha_i]),$$

Lemma 10.5 will follow from the equation

$$\varphi_* \sum_{i=1}^{4} [\alpha_i] = -\sigma,$$

$$\varphi_* \sum_{i=1}^{4} [\gamma_i] = -\tau \begin{bmatrix} & & 1 \\ & 1 & \\ -1 & & \end{bmatrix}.$$

Now $\sigma = u \otimes 1 + u \otimes f_2 + u \otimes f_3 + u \otimes f_4$ where $u \in H_1 K_0$ is defined by the matrix

$$\begin{bmatrix} 1 & & \\ & 1 & \\ & 3 & 1 \end{bmatrix}.$$

By the definition of $\varphi | \alpha_i$, we have

$$\varphi_*[\alpha_1] = -u \otimes 1$$

$$\varphi_*[\alpha_2] = -u \otimes f_2$$

since

$$f_2^{-1} \begin{bmatrix} 1 & & \\ & 1 & \\ & -3 & 1 \end{bmatrix} f_2 = \begin{bmatrix} 1 & & \\ & -2 & 3 \\ & -3 & 4 \end{bmatrix}.$$

Similarly, $\varphi_*[\alpha_3] = 1 \otimes f_3$ and $\varphi_*[\alpha_4] = 1 \oplus f_4$.

In the same way, we show that

$$\varphi_* \sum_{i=1}^{4} [\gamma_i] = \tau \begin{bmatrix} & & 1 \\ & 1 & \\ -1 & & \end{bmatrix}.$$

The reason for the factor

$$g = \begin{bmatrix} & & 1 \\ & 1 & \\ -1 & & \end{bmatrix}$$

is τ was defined in terms of the plane h_0 and g takes the plane h into h_0.

11. The Proof of Proposition 7.3 (concluded)

It was shown in the previous section that the composite

$$\mathbf{L} \to H_2 \, \partial M \to \mathbf{K} \tag{11.1}$$

is injective with cokernel at most $\mathbf{Z}_2$. The proof of Proposition 7.3 will be complete if we can show that this mapping is not surjective.

Suppose, on the contrary that the composite (11.1) is surjective. It follows immediately from diagram 7.2 and Lemma 10.4 that

$$H_2\Gamma \simeq H_2 M \simeq \mathbf{A} \oplus \mathbf{B}$$

as $SL(3;\mathbf{F}_3)$-modules. Furthermore, if α is the automorphism of $SL(3;\mathbf{Z})$ taking any matrix into its transpose inverse, then $\alpha\Gamma = \Gamma$. Since α interchanges the two conjugacy classes of maximal parabolic subgroups of $SL(3;\mathbf{Z})$, it is easy to see that

$$\alpha_*\colon H_2\Gamma \to H_2\Gamma$$

interchanges **A** and **B**.

Consider the spectral sequence of the extension

$$1 \to \Gamma \to SL(3;\mathbf{Z}) \to SL(3;\mathbf{F}_3) \to 1.$$

Then

$$E^2_{2,} \simeq H_2(SL(3;\mathbf{F}_3)) = 0$$

by van der Kallen [14], and

$$E^2_{1,1} \simeq H_1(SL(3,\mathbf{F}_3); H_1\Gamma)$$

contains only 3-torsion by Theorem 1.1. In addition, under the supposition above,

$$\begin{aligned} E^2_{0,2} &\simeq H_0(SL(3;\mathbf{F}_3), H_2\Gamma) \\ &\simeq H_0(SL(3;\mathbf{F}_3), \mathbf{A}\oplus\mathbf{B}) \\ &\simeq H_0(SL(3;\mathbf{F}_3), A)\oplus H_2(SL(3;\mathbf{F}_3), \mathbf{B}) \\ &\simeq \mathbf{Z}_2 \oplus \mathbf{Z}_2 \end{aligned}$$

and α_* interchanges these two summands.

On the other hand, van der Kallen [14] has shown that

$$H_2(SL(3;\mathbf{Z})) \simeq \mathbf{Z}_2 \oplus \mathbf{Z}_2$$

and it follows from his proof that the automorphism α induces the identity automorphism of this group. However this is clearly inconsistent with the spectral sequence above and the fact that

$$\alpha_*\colon E^2_{0,2} \to E^2_{0,2}$$

is nontrivial.

12. The Module $H^2\Gamma$

We begin by proving that the sequence (7.3) is exact.

Lemma 12.1. *Let J be the subgroup of scalar matrices in $\mathcal{M}(3;3)$. Then*

$$\mathrm{Ext}(\mathcal{M}_0(3;3);\mathbf{Z}) \simeq \mathcal{M}(3;3)/J.$$

Proof. By the definition of the functor Ext, we have an exact sequence

$$0 \to \mathrm{Hom}(\mathscr{M}_0; \mathbf{Z}) \to \mathrm{Hom}(\mathscr{M}_0; \mathbf{Q}) \to \mathrm{Hom}(\mathscr{M}_0; \mathbf{Q}/\mathbf{Z})$$
$$\to \mathrm{Ext}(\mathscr{M}_0; \mathbf{Z}) \to \mathrm{Ext}(\mathscr{M}_0; \mathbf{Q})$$

where $\mathscr{M}_0 = \mathscr{M}_0(3;3)$. Since $\mathrm{Hom}(\mathscr{M}_0; \mathbf{Q}) = 0$ and $\mathrm{Ext}(\mathscr{M}_0; \mathbf{Q}) = 0$,

$$\mathrm{Ext}(\mathscr{M}_0; \mathbf{Z}) \simeq \mathrm{Hom}(\mathscr{M}_0; \mathbf{Q}/\mathbf{Z})$$
$$\simeq \mathrm{Hom}(\mathscr{M}_0; \mathbf{F}_3).$$

Let $t\colon \mathscr{M}(3;3) \to \mathbf{F}_3$ be the trace homomorphisms so that

$$0 \to \mathscr{M}_0 \to \mathscr{M}(3;3) \xrightarrow{t} \mathbf{F}_3 \to 0$$

is an exact sequence of $\overline{SL}$-modules, $\overline{SL}$ acting trivially on $\mathbf{F}_3$. Passing to the dual vector spaces, we have the exact sequence

$$0 \to \mathrm{Hom}(\mathbf{F}_3, \mathbf{F}_3) \to \mathrm{Hom}(\mathscr{M}(3;3); \mathbf{F}_3) \to \mathrm{Hom}(\mathscr{M}_0; \mathbf{F}_3) \to 0.$$

Now $\mathscr{M}(3;3)$ is easily seen to be a self dual $\overline{SL}$-module and $\mathrm{Hom}(\mathbf{F}_3, \mathbf{F}_3)$ is a trivial $\overline{SL}$-module. It is easy to see that J is the only subgroup of $\mathscr{M}(3;3)$ with trivial $\overline{SL}$ action so Lemma 12.1 follows.

Note that $t\colon \mathscr{M}(3;3) \to \mathbf{F}_3$ takes J into zero so we have

$$t\colon \mathscr{M}(3;3)/J \to \mathbf{F}_3.$$

Lemma 12.2. *The mapping*

$$t_*\colon H_0(\overline{SL}; \mathscr{M}(3;3)/J) \to H_0(\overline{SL}; \mathbf{F}_3) \simeq \mathbf{F}_3$$

is an isomorphism.

Proof. Consider the commutative diagram

$$\begin{array}{ccccc} H_0(\overline{SL}; \mathscr{M}(3;3)) & \longrightarrow & H_0(\overline{SL}; \mathscr{M}(3;3)/J) & \to & 0 \\ & {\scriptstyle t_*}\searrow \quad \swarrow{\scriptstyle t_*} & & & \\ & H_0(SL; \mathbf{F}_3) & & & \end{array}$$

By definition, $H_0(\overline{SL}; \mathscr{M}(3;3)) = \mathscr{M}(3;3)/I$ where I is the subgroup of $\mathscr{M}(3;3)$ generated by elements of the form $A - A \cdot g$, $A \in \mathscr{M}(3;3)$, $g \in \overline{SL}$. It is easy to see from this that $t\colon \mathscr{M}(3;3) \to \mathbf{F}_3$ defines an isomorphism from $H_0(\overline{SL}; \mathscr{M}(3;3))$ onto $\mathbf{F}_3$ and the lemma then follows from the diagram above.

Recall that $\mathbf{L}$ is the kernel of the homomorphism $H_2\, \partial M \to H_2 M$ induced by inclusion and that $\mathrm{tor}\, L \simeq \mathscr{M}(3;3)/J$. Thus we have a short exact sequence

$$0 \to \mathrm{tor}\, \mathbf{L} \to \mathbf{L} \to \mathbf{L}/\mathrm{tor}\, \mathbf{L} \to 0$$

and the resulting exact sequence

$$\cdots \to H_0(\overline{SL}; \mathrm{tor}\, \mathbf{L}) \xrightarrow{\eta} H_0(\overline{SL}; \mathbf{L}) \to H_0(\overline{SL}; \mathbf{L}/\mathrm{tor}\, \mathbf{L}) \to 0. \tag{12.1}$$

Let $\mu \in \mathbf{L}$ be defined by

$$\mu=(\sigma+x)\begin{bmatrix}1 & & \\ & 1 & 1\\ & & 1\end{bmatrix}-(\sigma+x)\begin{bmatrix}1 & & \\ & 1 & 1\\ & -1 & \end{bmatrix}+(\sigma+x)\begin{bmatrix}1 & & \\ & 1 & -1\\ & 1 & \end{bmatrix}-(\sigma+x)\begin{bmatrix}1 & & \\ & 1 & -1\\ & & 1\end{bmatrix}$$

where σ and x are defined in Section 9. Clearly μ is a sum of elements of the form $A-Ag$ so it represents zero in $H_0(\overline{SL}; \mathbf{L})$. A tedious but straightforward computation shows that μ is in fact in tor $\mathbf{K}$ and given by the expression

$$\mu=\alpha\begin{bmatrix}1 & 1 & \\ & 1 & \\ & & 1\end{bmatrix}-\alpha\begin{bmatrix}1 & -1 & \\ & 1 & \\ & & 1\end{bmatrix}-(\alpha-\beta)\begin{bmatrix}1 & -1 & -1\\ & 1 & \\ & & 1\end{bmatrix}+(\alpha-\beta)\begin{bmatrix}1 & 1 & 1\\ & 1 & \\ & & 1\end{bmatrix}$$

$$+(\alpha+\beta)\begin{bmatrix}1 & 1 & -1\\ & 1 & \\ & & 1\end{bmatrix}-(\alpha+\beta)\begin{bmatrix}1 & -1 & 1\\ & 1 & \\ & & 1\end{bmatrix}-\beta\begin{bmatrix}1 & & 1\\ & 1 & \\ & & 1\end{bmatrix}+\beta\begin{bmatrix}1 & & -1\\ & 1 & \\ & & 1\end{bmatrix}$$

where α and β are defined in Section 9. Recall that α and β are elements in tor $H_1(K_0 \times_{\bar{P}_0} \bar{P}_1)$ which map to zero in $H_1 K_1$. Since the eight matrices appearing in the expression for μ above lie in different cosets of $\bar{P}_0 \backslash \bar{P}_1$, the element μ is a non-zero torsion element in

$$H_1(K_0 \times_{\bar{P}_0} \overline{SL}) \simeq H_1(K_0 \times_{\bar{P}_0} \bar{P}_1) \otimes_{\Lambda_1} \Lambda.$$

In fact, for any $D \in SL(2; \mathbf{F}_3)$,

$$\mu\begin{bmatrix}1 & \\ & D\end{bmatrix}=\mu.$$

This is proved by verifying it when D is one of the two matrices

$$\begin{bmatrix} & -1\\ 1 & \end{bmatrix}, \quad \begin{bmatrix}1 & 1\\ & 1\end{bmatrix}$$

since $SL(2; \mathbf{F}_3)$ is generated by these two matrices. Now, under the identification of tor $\mathbf{K}$ with $\mathscr{M}(3; 3)/J$, μ corresponds to a non-zero element invariant under the action of $SL(2; \mathbf{F}_3)$.

Lemma 12.3. *The only elements of $\mathscr{M}(3; 3)/J$ invariant under $SL(2; \mathbf{F}_3)$ are represented by*

$$\begin{bmatrix}0 & & \\ & a & \\ & & a\end{bmatrix}, \quad a \in \mathbf{F}_3.$$

Proof. If $D \in \mathscr{M}(3, 3)/J$ is invariant under

$$\begin{bmatrix}1 & & \\ & -1 & \\ & & -1\end{bmatrix},$$

then, mod J, D has the form

$$\begin{bmatrix} 0 & 0 & 0 \\ 0 & a & b \\ 0 & c & d \end{bmatrix}.$$

The result now follows easily.

We summarize the properties of the element μ in the following.

Lemma 12.4. *The element μ in* tor $\mathbf{K}$ *defines a non-zero element in* $\ker \gamma$ *where* $\gamma\colon H_0(\overline{SL};\ \text{tor}\ \mathbf{K}) \to H_0(\overline{SL};\ \mathbf{K})$ *is the homomorphism of the exact sequence* (12.1).

Proof. As a consequence of Lemma 12.3, we know that μ represents one of the following two elements in $\mathscr{M}(3;3)/J$:

$$\begin{bmatrix} 0 & & \\ & 1 & \\ & & 1 \end{bmatrix} \quad \text{or} \quad \begin{bmatrix} 0 & & \\ & -1 & \\ & & -1 \end{bmatrix}.$$

Since both of these elements have trace non-zero, it follows from Lemma 12.2 that μ represents a non-zero element in $H_0(\overline{SL};\ \text{tor}\ \mathbf{K})$. As we observed earlier, μ is the sum of elements of the form $A - Ag$ so $\gamma(\mu) = 0$.

According to Lemma 12.2, $H_0(\overline{SL};\ \text{tor}\ \mathbf{K}) \simeq \mathbf{F}_3$ so Lemma 12.4 implies that $\gamma\colon H_0(\overline{SL};\ \text{tor}\ \mathbf{K}) \to H_0(\overline{SL};\ \mathbf{K})$ is the zero homomorphism. As an immediate consequence, we have the

Corollary. *The sequence*

$$0 \to \mathscr{M}(3;3)/J \to H^2\,\Gamma \to \text{Hom}(H_2\Gamma;\ \mathbf{Z}) \to 0$$

does not split over $\Lambda = \mathbf{Z}[SL]$.

The following result is needed in [9].

Lemma 12.5. *The group $H_0(\overline{SL};\ H^2\,\Gamma)$ is cyclic of order two.*

Proof. By Proposition 7.4, we know that $H^2\,\Gamma \simeq H^2\,M$ is isomorphic to $\mathbf{L}$. Furthermore, since the homomorphism η in the exact sequence (12.1) is zero, we know that

$$H_0(\overline{SL};\ \mathbf{L}) \simeq H_0(\overline{SL};\ \mathbf{L}/\text{tor}\ \mathbf{L}).$$

Now, by the results of Section 7, we have the exact sequence

$$0 \to \mathbf{L}/\text{tor}\ \mathbf{L} \to \mathbf{A} \oplus \mathbf{B} \to \mathbf{Z}_2 \to 0$$

and therefore the exact sequence

$$0 \to H_0(\overline{SL};\ \mathbf{L}/\text{tor}\ \mathbf{L}) \to H_0(\overline{SL};\ \mathbf{A} \oplus \mathbf{B}) \to H_0(\overline{SL};\ \mathbf{Z}_2) \to 0.$$

Since $\mathbf{A}$, $\mathbf{B}$ are $\overline{SL}$-modules defined by inducing up from non-trivial one dimensional representations of subgroups, it follows easily that

$$H_0(\overline{SL};\ \mathbf{A}) \simeq H_0(\overline{SL};\ \mathbf{B}) \simeq \mathbf{Z}_2$$

and Lemma 12.5 is proved.

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